



Deep transfer learning for image classification: a survey

Jo Plested¹ · Musa Phiri¹ · Tom Gedeon^{2,3,4}

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Abstract

Deep neural networks such as convolutional neural networks (CNNs) and transformers have achieved many successes in image classification in recent years. It has been consistently demonstrated that best practice for image classification is when large deep models can be trained on abundant labeled data. However, there are many real world scenarios where the requirement for large amounts of training data to get the best performance cannot be met. In these scenarios, transfer learning can help improve performance. To date, there have been no surveys that comprehensively review deep transfer learning as it relates to image classification overall. We believe it is important for future progress in the field that current knowledge is collated and the overarching patterns analyzed and discussed. In this survey we formally define deep transfer learning and the problem it attempts to solve in relation to image classification. We survey the current state of the field and identify where recent progress has been made. We show where the gaps in current knowledge are and make suggestions for how to progress the field to fill in these knowledge gaps. We present new taxonomies of the solution and applications of transfer learning for image classification. These taxonomies make it easier to see overarching patterns of where transfer learning has been effective and, where it has failed to fulfil its potential. This also allows us to suggest where the problems lie and how it could be used more effectively. We demonstrate that under this new taxonomy, many of the applications where transfer learning has been shown to be ineffective or even hinder performance are to be expected when taking into account the source and target datasets and the techniques used. In many of these cases, the key problem is that methods and hyperparameter settings designed for large and very similar target datasets are used for smaller and much less similar target datasets. We identify alternative choices that could lead to better outcomes.

Keywords Deep transfer learning · Image classification · Convolutional neural networks · Deep learning

1 Introduction

It has been consistently shown that the success of deep neural networks for image classification relies upon an abundance of training data (Ngiam et al. 2018; Mahajan et al. 2018; Kolesnikov et al. 2020; Cherti et al. 2023). However, there are many real world scenarios where this requirement cannot be met. These include:

1. Insufficient data because the data is very rare or there are issues with privacy etc. This scenario is common in the medical image domain (Kora et al. 2022).
2. It is prohibitively expensive to collect and/or label data. This is common for image tasks related to robotics. While large amounts of data can be created in simulation, when the model is transferred to real-world robots, it must be fine-tuned on limited data. This is because many robotics devices are expensive, and data needs to be collected in real time (Jaquier et al. 2025).
3. The long tail distribution where a small number of classes are very frequent and thus easy to model, while many more are rare and harder to model (Bengio 2015). This is common with self-driving cars where usual driving conditions are well represented, but emergency type situations are rare (Wang et al. 2021).

There may be restraints on compute resources that limit training a large model from random initialization with large amounts of data. For example, environmental concerns (Strubell et al. 2020).

In all these scenarios transfer learning has the potential to significantly improve performance. In this paradigm the model is pretrained on a related dataset and task for which more data is available or, more recently, a more general task on the same dataset (He et al. 2022; El-Nouby et al. 2021). For this process to improve rather than harm performance the dataset must be related closely enough and best practice methods used.

With the exponentially increasing demand for the application of modern deep neural networks to a wider array of real world areas, work in transfer learning has increased at a commensurate pace making it important to regularly take stock. In this paper we survey the current state of the field, where recent progress has been made and where there are gaps in current knowledge. We also make suggestions for how to progress the field to fill in these knowledge gaps. While there are many surveys in related domains and specific sub areas, to the best of our knowledge there are none that focus on deep transfer learning for image classification in general. We believe it is important for future progress in the field that current knowledge is collated and the overarching patterns analyzed and discussed.

We make the following contributions:

1. Formally defining and categorizing deep transfer learning (DTL) and the problem it attempts to solve as it relates to image classification. Previous reviews have provided informal definitions of (DTL), and taxonomies (Tan et al. 2018; Iman et al. 2023; Fuchao et al. 2022). We refine these definitions and extend the taxonomies so they are more suited to modern (DTL) solutions and their application to image classification.
2. Performing a thorough review and analysis of recent progress in the field.
3. Presenting a continuous transition taxonomy highlighting the continuous nature of the source and target dataset relationships in transfer learning applications and their effect

- on the optimal solutions. This taxonomy helps us draw attention to the failure of transfer learning solutions to perform as expected in certain application areas.
4. A new way of considering negative transfer so it can be eliminated in all scenarios.
 5. Giving a detailed summary of source and target datasets commonly used in the area to help the reader find the ideal source dataset to match their target task.
 6. Summarizing current knowledge in the area as well as pointing out knowledge gaps and suggested directions for future research.

In Sect. 2 we discuss the relationship between this survey and others in the area from general transfer learning to more closely related domains. In Sect. 3 we introduce the problem domain and formalize the difficulties with learning from small datasets that transfer learning attempts to solve. This section includes terminology, definitions, and our new taxonomies that are referred to throughout this paper. Sections 4 to 7 summarize current knowledge in relation to each of our five solution categories in our new taxonomy. Finally, Sect. 8 summarizes the overarching themes of these solutions, and Sect. 9 summarizes our paper in full and recommends next steps for the field.

2 Related work

Many reviews related to DTL have been published in the past decade. The similarities and differences are summarized in Table 1. The existing reviews differ from ours in two important ways. The first group consists of more general reviews that provide a high level overview of transfer learning and attempt to include all machine learning and task sub-fields. Reviews in this group are shown under the heading "Broad" in Table 1. The second group is more specific, with reviews providing a comprehensive breakdown of the progress on a particular, narrow, domain specific task. They are shown under the heading "Narrow" in Table 1 and discussed further in the relevant parts of Sects. 4 and 7. Our review is designed to sit in between these two groups and provides an overarching review of DTL as it relates to image classification specifically so that overarching patterns can be identified, analyzed and discussed. This allows us to make recommendations for best practice when using deep transfer learning for image classification (DTLIC), as well as identify knowledge gaps and suggest directions for future research. A few surveys seem more closely related to ours, but they are not full reviews, only briefly introduce the area of DTLIC, and do not analyze recent advances in the field. These reviews are covered under the heading "Closely related" in Table 1.

3 Overview

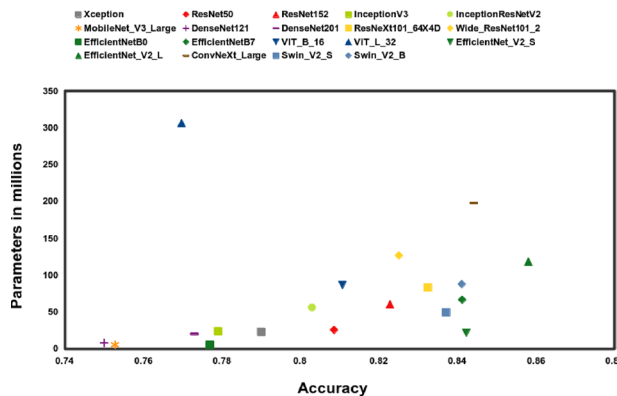
Image classification tasks model relationships in complex hierarchical data. For example, there is no obvious relationship between raw RGB pixel values and whether the image is of a cat or a dog, and it would be impossible to hand-design a transformation that separates the two classes. For this reason, a large deep learning model is generally needed for image classification tasks, with millions or even billions of parameters arranged in layers that learn hierarchical combinations of features. The more complex the relationship between the input

Table 1 Transfer learning survey papers

Paper	Year	Focus	Deep learning focus	Image classification focus	Taxonomy	Full review
Closely related						
Panda et al. (2024)	2024	DTLIC	Yes	Yes	No	No
Ribani and Marengoni (2019)	2019	DTLIC	Yes	Yes	No	No
Broad						
Iman et al. (2023)	2023	DTL	Yes	No	Yes	Yes
Fuchao et al. (2022)	2022	DTL	Yes	No	Yes	Yes
Tan et al. (2018)	2018	DTL	Yes	No	Yes	Yes
Narrow						
Ismail et al. (2025)	2025	Agriculture	Yes	No	NA	Limited
Jaquier et al. (2025)	2025	Robotics	Yes	No	NA	Limited
Kora et al. (2022)	2022	Medical images	Yes	Limited	NA	Limited
Li and Deng (2020)	2020	Facial expression recognition	Yes	Limited	NA	Limited
Minaee et al. (2022)	2020	Image segmentation	Yes	No	NA	Limited
Mazurowski et al. (2019)	2019	Radiology	Yes	Limited	NA	Limited
Ghosh et al. (2019)	2019	Image segmentation	Yes	No	NA	Limited

Closely related papers are ones that sound most similar to ours based on the title but are limited in some way, broad are reviews of transfer learning in general not limited to image classification, and narrow are focused on a particular application of transfer learning. Under the narrow category, taxonomies are listed as NA for not applicable where their scope is limited, so any taxonomy would not be relevant to this work. For the same reason, the review is listed as limited even if it is a full review if it has a narrow scope, thus limited crossover with the scope of this paper. DTLIC refers to deep transfer learning for image classification

Fig. 1 Increase in performance on ImageNet 1 K due to model size, measured by number of parameters in millions. In general, larger models result in better performance. However, there are outliers that show a decrease in performance for the largest models. This is because the largest models overfit ImageNet1k



data and the prediction task, the more learnable parameters are needed to approximate the transformation function.

An example of complex hierarchical data is ImageNet 1k. Figure 1 shows that, in general, the performance on ImageNet 1K (Deng et al. 2009) increases with the number of

model parameters. However, some outliers show a decrease in performance as the model continues to increase in size. This is because the largest deep learning models overfit even the 1.3 million training examples in ImageNet1K (Ngiam et al. 2018; Mahajan et al. 2018; Kolesnikov et al. 2020; Liu et al. 2022).

3.1 Learning from small vs large datasets

When fitting large models to small, complex data, as is often needed with real-world data, it is essential to consider the classic *bias-variance trade-off*. We are attempting to minimize the error, which is the difference between the predictions of our model and the true output values. This difference is defined by a performance measure P and the loss we are attempting to minimize L . We define f^* as the true underlying function we are approximating. The error in our approximation originates from two places:

1. *Bias* ε_{bias} : the function that can be approximated is restricted by the choices we make regarding the model architecture. We define $\mathcal{F}$ as the set of functions defined by our model architecture and $f_{\mathcal{F}}^*$ as the function in $\mathcal{F}$ that minimizes our loss. This is unlikely to be the exact underlying function f^* , so our bias error becomes the difference between the optimal function in $\mathcal{F}$ and the true underlying function. We define this error as ε_{bias} . Our bias error will be larger the more restrictive our model architecture is relative to the underlying function we are approximating, f^* . For example, if we try to approximate a higher order polynomial function like x^2 with a linear model x we will have a high bias.
2. *Variance* ε_{var} : the difference between $f_{\mathcal{F}}^*$ and the function that minimizes the loss from a particular set of training data. We define $L(f_n)$ as the empirical estimate of the loss given a particular dataset. So, our variance error becomes the difference between minimizing the empirical loss and the true loss for the optimal function in $\mathcal{F}$. We define this error as ε_{var} . If we have a small number of examples in our training dataset and are attempting to model a complex function, our training data is unlikely to fully capture the function's complexities.

The total loss is then:

$$L = \varepsilon_{bias} + \varepsilon_{var} \quad (1)$$

being the sum of the bias loss coming from restrictions due to our choice in model architecture, and the variance loss coming from minimizing the empirical loss, based on the samples in our dataset, rather than the true underlying loss based on an infinite sample of all possible data and labels.

It should be noted that the above errors depend entirely on the loss function we define. If our loss function does not accurately reflect the task we are trying to learn, this is a value alignment problem (Gabriel 2020), which is beyond the scope of this discussion.

The bias error ε_{bias} measures how closely functions in $\mathcal{F}$ can approximate the optimal solution f^* . The variance error ε_{var} measures the effect of minimizing the empirical loss $L(f_n)$ instead of the true underlying loss $L(f^*)$ (Bottou and Bousquet 2008). So finding a function that is as close as possible to f^* can be broken down into:

1. Choosing a class of models that is more likely to contain the optimal function
2. Choosing training examples that better approximate an infinite set of all possible data and labels.

In general, ε_{var} can be reduced by having a larger number of examples (Wang et al. 2020). Thus, when there are sufficient and varied labeled training examples in our training dataset D_{train} , the empirical loss $L(f_n)$ can provide a good approximation of the true underlying loss $L(f_{\mathcal{F}}^*)$ the optimal f in $\mathcal{F}$. When n the number of training examples is small the empirical loss may be further from being a good approximation of the expected loss. The resultant empirical loss minimizer overfits. For example, Ribeiro et al. created a curated dataset of images of wolves and huskies, ensuring that all wolves are on snow-covered backgrounds and all huskies are on snow-free backgrounds (Ribeiro et al. 2016). They showed that a neural network classifier trained on this dataset to predict whether an image shows a wolf or a husky focuses solely on the background, not on the animal itself. This is an extreme example of a model overfitting the idiosyncrasies of the dataset by minimizing the empirical loss, which is not a good approximation of the true underlying loss.

To alleviate the problem of having an unreliable empirical loss when D_{train} is not sufficient, prior knowledge can be used to augment the data in D_{train} , constrain the candidate functions $\mathcal{F}$, or constrain the parameters of f via initialization or regularization (Wang et al. 2020).

In this work, we focus on transfer learning as a form of constraining the parameters of f to address the unreliable empirical loss problem. When weights pretrained on a large related dataset are used to initialize a deep neural network instead of random initialization, the loss landscape tends to be flatter and easier to navigate. This means that complex models with large numbers of trainable parameters are less likely to overfit to sharp local minima corresponding to idiosyncrasies of the small target dataset (Liu et al. 2019). Section 3.8 briefly discusses how deep transfer learning relates to other techniques that use prior knowledge to solve the small dataset problem.

3.2 Deep transfer learning

DTL is transfer learning applied to deep neural networks. Much of the widely accepted terminology used to describe transfer learning comes from an early review (Pan and Yang 2009). The authors define transfer learning as:

Definition 1 Given a source domain $\mathcal{D}_S$ and learning task T_S , a target domain $\mathcal{D}_T$ and learning task T_T , transfer learning aims to help improve the learning of the target predictive function $f_T(\cdot)$ in $\mathcal{D}_T$ using the knowledge in $\mathcal{D}_S$ and T_S , where $\mathcal{D}_S \neq \mathcal{D}_T$, or $T_S \neq T_T$.

Tan et al. (Tan et al. 2018) define DTL as follows: Given a transfer learning task defined by $\mathcal{D}_S, \mathcal{D}_T, T_S, T_T$ and $f_T(\cdot)$. It is a deep transfer learning task where $f_T(\cdot)$ is a non-linear function that reflected a deep neural network.

For the purposes of this paper we refine this definition to be more precise:

Definition 2 A transfer learning task is defined by $\mathcal{D}_S, \mathcal{D}_T, T_T$ and an optimal target function $f_T(\cdot)$. It is a deep transfer learning task if 1) $f_T(\cdot)$ is approximated by a deep neural

network and 2) a deep neural network $f_S(\cdot)$ trained on the source dataset is used to improve the approximation of the target function $f_T(\cdot)$.

We define the source model as the deep neural network trained on the source dataset, the target model as the deep neural network trained on the target dataset, and the pretrained weights as the values of the weights after training on the source dataset is complete. Usually, DTL involves initializing some or all of the weights of the target model with the pretrained weights from the source model. If the source and target tasks are not identical then the final prediction layer weights must be randomly initialized rather than transferring them from the source model as the pretrained values will not be appropriate for the target task.

Fine-tuning the weights is defined as continuing to train any transferred weights on the target task.

Freezing the weights is defined as only training the newly initialized weights on the target task.

Any reinitialized weights must be trained. Figure 2 shows the pretraining and fine-tuning pipeline of DTL.

Combining the discussion from Sect. 3.1 with definition 2 using deep transfer learning techniques to pretrain weights W can be thought of as regularizing W . Initializing W with weights that have been well-trained on a large source dataset rather than with very small random values makes it more difficult for the weights to move far from their pretrained values (Neyshabur et al. 2020; Liu et al. 2019). In the most effective applications of DTL the source dataset is many orders of magnitude larger than the target dataset. One example is pretraining on ImageNet 1K with 1.3 million training images and transferring to medi-

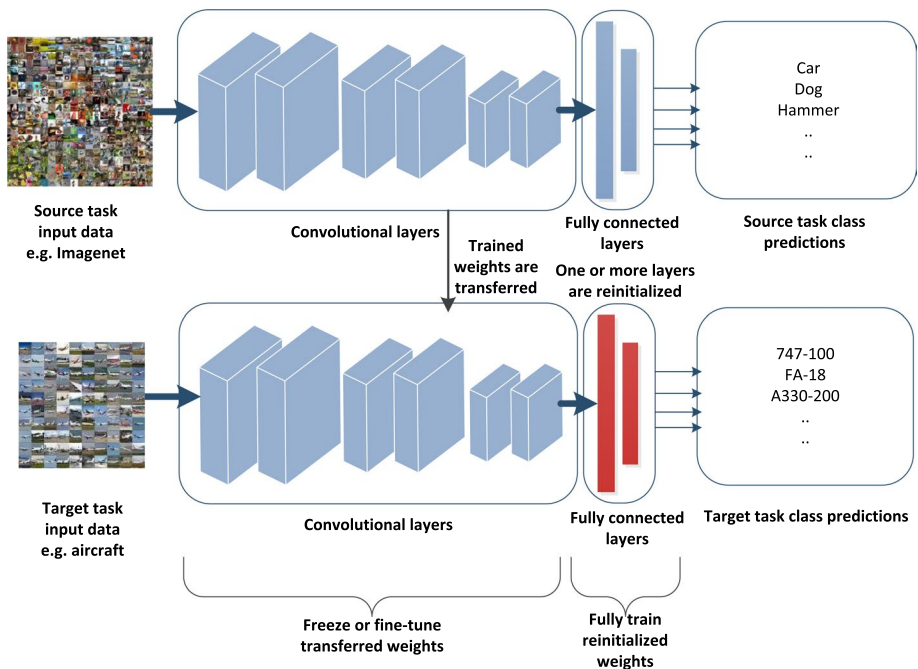


Fig. 2 Deep transfer learning

cal imaging tasks which often only have 100s of labeled examples. So even with the same learning rate and number of epochs, the number of updates to the weights while training on the target dataset will be orders of magnitude fewer than for pretraining. When done well this prevents the model from learning weights that are based on noise or idiosyncrasies in the small target dataset. When done poorly this can prevent the model from effectively learning features unique to the target dataset that could improve the model's performance. In the worst case scenario, the latter can result in negative transfer, which we describe in detail in Sect. 3.5.

3.2.1 Analysis of the transfer learning optimization space

Several feature space analysis methods were used in (Neyshabur et al. 2020). They found that models trained from pretrained weights make similar mistakes on the target domain, have similar features and are surprisingly close in ℓ_2 distance in the parameter space. They are in the same basins of the loss landscape. Models trained from random initialization do not live in the same basin, make different mistakes, have different features and are farther away in ℓ_2 distance in the parameter space.

A flatter and easier to navigate loss landscape for pretrained models compared to their randomly initialized counterparts was also shown in (Liu et al. 2019). They showed improved Lipschitzness and that this accelerates and stabilizes training substantially. Particularly that the singular vectors of the weight gradient with large singular values are shrunk in the weight matrices. Thus, the magnitude of the gradient back-propagated through a pretrained layer is controlled and pretrained weight matrices stabilize the magnitude of the gradient especially in lower layers, leading to more stable training.

The above results mean that a pretrained model is likely to perform well and lead to more stable training when the source and target datasets are closely related. However, this stability means that it is more difficult to change the weights when they are not suited to the target task. This means that a much higher learning rate is usually needed (Dai et al. 2024).

3.3 Learning from similar vs dissimilar datasets

In transfer learning it is important to match the source and target datasets as closely as possible. A more closely matched unlabeled source dataset will usually perform better than a less closely matched labeled dataset (Zoph et al. 2020; Azizi et al. 2021).

Dataset similarity refers to the quantifiable degree of shared characteristics between source and target datasets (Torrey and Shavlik 2010; Zhuang et al. 2020). The transferability of learned representations in deep neural networks is heavily influenced by dataset similarity, as greater similarity often enables more effective reuse of features. In contrast, large dissimilarities can lead to negative transfer, which is discussed in Sect. 3.5. Similarity can be quantified through a similarity function $S : \mathcal{D} \times \mathcal{D} \rightarrow \mathbb{R}$, where $\mathcal{D}$ represents the space of all possible datasets, and $S(d_1, d_2)$ yields a bounded, symmetric score indicating closeness between datasets (Ben-David et al. 2010; Rozinek and Mareš 2021).

As noted by Stolte et al. (Stolte et al. 2024), defining “similarity” is challenging when datasets differ in size, dimensionality, feature space, or label set. In some cases, source and target domain similarity can be effectively estimated through intuition and/or similarity between class labels (Mahajan et al. 2018; Ngiam et al. 2018; Puigcerver et al. 2020).

When this is not possible or we would like a more robust measure, there are many ways of measuring domain similarity. Gupta et al. (Gupta et al. 2025) group similarity measurement techniques into four categories:

1. *Correlation-based metrics*, such as the Pearson, Spearman, Kendall rank, and point-biserial correlation coefficients, quantify statistical associations between features.
2. *Distance-based metrics*, including Euclidean, Manhattan, Mahalanobis, Canberra, Levenshtein, Wasserstein, Fréchet distance (FD), Earth Mover's Distance (EMD), and Hamming distances, assess geometric differences in feature space.
3. *Probability-based measures*, such as mutual information, conditional probability, and divergence metrics including Kullback–Leibler, Jensen–Shannon, Bhattacharyya, Maximum Mean Discrepancy (MMD), Hellinger, and total variation distance, compare underlying data distributions.
4. *Set-based measures*, such as the Jaccard Index, Dice Coefficient, Cosine Similarity, and Overlap Coefficient, evaluate overlap in categorical or discrete representations.

These measures provide complementary perspectives, and the choice of method depends on the data modality, domain, and intended transfer task.

In practice, the suitability of a similarity measure depends strongly on dataset characteristics and the nature of the domain gap.

MMD has proven especially effective for aligning high-dimensional visual domains with distribution shifts but similar underlying semantics, successfully enabling knowledge transfer between medical imaging modalities such as CT and MRI scans (Long et al. 2015; Chen et al. 2019). For scenarios requiring optimal transport solutions, such as few-shot settings where cluttered backgrounds, large intra-class variations and the loss of local structural information make global embeddings unreliable, EMD excels by comparing structured representations through optimal matching (Zhang et al. 2020). This approach has demonstrated strong performance on challenging datasets such as miniImageNet and tieredImageNet (Zhang et al. 2020). Wasserstein distance proves particularly valuable for applications where source and target domains exhibit minimal distributional overlap, such as transferring knowledge from simulation data to real sensor measurements in fault diagnosis systems (Cheng et al. 2019). When computational efficiency is paramount, Bhattacharyya distance provides effective similarity estimation without requiring expensive fine-tuning procedures, making it ideal for rapid model selection in computer vision applications (So and Chung 2017).

These examples highlight how data modality, degree of semantic alignment, and size of the domain gap inform the optimal choice of similarity measure for deep transfer learning.

3.4 Example showing the effects of target dataset size and similarity on deep transfer learning

To motivate our discussions on the importance of considering the size of the target dataset and its similarity to the source dataset in determining the optimal transfer learning solution for a given target task we present a case study where deep transfer learning fine-tuning hyperparameters designed for moderately sized target datasets that are externally similar to the target dataset perform poorly on smaller, dissimilar target datasets.

3.4.1 Big transfer-transferring from larger source datasets to small, similar target datasets

Big Transfer (BiT) was defined as pretraining on source datasets larger than ImageNet 1K in (Kolesnikov et al. 2020). The authors devise general fine-tuning hyperparameter rules that are only dependent on the size of the target dataset. They show that these rules result in close to optimal performance for a range of datasets that are either extremely similar to the source datasets or, in the case of Flowers, known to respond well to protocols designed for extremely similar target datasets. They also show that decaying the learning rate faster works better for moderately sized compared to large target datasets. We hypothesise that these best practice recommendations will be far from optimal for less closely related target datasets, as has been shown with many other methods and recommendations (Xuhong et al. 2018; Li et al. 2021; Wan et al. 2019; Plested et al. 2021; Li et al. 2019; Chen et al. 2019; Kou et al. 2020).

3.4.2 Experiment extending big transfer to less similar datasets

An extension to the Big Transfer experiments was performed to show how training with hyperparameters recommended for small, target datasets similar to the source dataset performs on two less similar target datasets. We chose two datasets that are known to be difficult to transfer to and conceptually different from our source dataset, ImageNet21K. These are Stanford Cars (Cars) (Kraus et al. 2017) and FGVC Aircraft (Aircraft) (Maji et al. 2013). We first measure the similarity of all target datasets to ImageNet21K using three measures, EMD (Zhang et al. 2020), Wasserstein distance (Cheng et al. 2019), and FD described in Sect. 3.3. The results of these measures are shown in Table 2 and show that, as expected, our two new datasets have much lower similarity measures than the original datasets.

We then trial different fine-tuning hyperparameters than those recommended for more similar target datasets. The results are shown in Table 3. It can be seen that the recommended learning rate is effective for the most similar datasets, being Cifar10, Cifar100, and Oxford Pets. However, it is far from optimal for the less similar datasets, being Aircraft and Cars. While the decay schedule does not make a significant difference to final mean accuracy, it does produce more robust results with far smaller standard deviations in both

Table 2 Dataset similarity and distance measures for five target datasets compared to ImageNet1k as the source dataset

Target datasets	EMD (distance)	EMD (similarity)	Wasserstein (distance)	Wasserstein (similarity)	FD (distance)	FD (similarity)
Cifar10	0.9755	0.6140	893.2167	0.6398	755.6660	0.6749
Cifar100	0.8783	0.6446	760.7420	0.6836	511.1466	0.7665
Oxford pets	1.0362	0.5956	1012.8434	0.6026	1238.4232	0.5250
Cars	1.1286	0.5688	1152.1820	0.5621	1323.2090	0.5024
Aircraft	1.2809	0.5270	1278.8186	0.5276	1415.3668	0.4788

The similarity measures are normalized so that 1.0 is identical and 0 is no similarity and distance scores are unnormalized with higher distances meaning the datasets are less similar. The measures show that the two datasets that are conceptually similar to ImageNet1K, being Cifar10 and Cifar100 have higher similarity scores and lower distance measures than those that are less conceptually similar, Cars and Aircraft, and known to be harder to transfer to from ImageNet1K

Table 3 Extension of big transfer (BiT) to less similar datasets (Kolesnikov et al. 2020)

Dataset	Recommended values	Less similar lr	Less similar decay	Less similar combined	Best result	Best hyperparameters
Cifar10	97.626 ± 0.108	97.402 ± 0.069	97.014 ± 0.019	97.086 ± 0.108	97.626 ± 0.108	lr = 0.003, lr decay steps = 100, rate = 0.1
Cifar100	86.93 ± 0.210	84.977 ± 0.224	86.484 ± 0.149	85.238 ± 0.263	86.93 ± 0.210	lr = 0.003, lr decay steps = 100, rate = 0.1
Oxford pets	92.856 ± 0.095	91.538 ± 0.806	92.914 ± 0.138	92.026 ± 0.082	92.856 ± 0.095	lr = 0.003, lr decay steps = 100, rate = 0.1
Cars	81.812 ± 0.151	87.602 ± 0.692	81.825 ± 0.216	87.512 ± 0.335	87.958 ± 0.239	lr = 0.025, lr decay steps = 5, rate = 0.96
Aircraft	82.414 ± 0.177	87.256 ± 4.750	81.448 ± 0.256	86.95 ± 0.308	86.95 ± 0.308	lr = 0.02, lr decay steps = 3, rate = 0.97

All results are obtained using a ResNet-50x1 pretrained on ImageNet-21k (BiT-M). Cifar10 (Krizhevsky 2009), Cifar100 (Krizhevsky 2009) and Oxford Pets (Parkhi et al. 2012) are examples of datasets from the original paper that are very similar to ImageNet-21k, while Cars (Krause et al. 2013) and Aircraft (Maji et al. 2013) are less similar. Recommended values are those values recommended for learning rate and learning rate decay from the paper. Less similar learning rate (lr) and less similar decay are the best lr (lr = 0.02) and lr decay (lr decay steps = 3, rate = 0.97) for both less similar datasets overall. Less similar combined is the combination of the best lr and lr decay for less similar datasets. Best result is the best result with any hyperparameters for the given dataset individually, taking into account both mean accuracy and standard deviation

cases. So, when looking for one best final prediction model, the decay schedule should be considered, whereas for an ensemble of models, it may be less important.

For the best and most robust results, the similarity of the source and target datasets, in addition to the size of the target dataset, should be taken into account when setting fine-tuning hyperparameters. It should be noted that the effect of using a learning rate suited to more similar target datasets on less similar target datasets is much more pronounced than vice versa. In general, small, less similar target datasets are more sensitive to transfer learning protocols.

3.5 Negative transfer

If the source dataset is not well related to the target dataset the target model can be negatively impacted by pretraining. This situation is called negative transfer.

Wang et al. elaborate on factors that affect negative transfer (Wang et al. 2019a, b):

- Similarity between the source and target domains. Transfer learning assumes that there

is some similarity between joint distributions in source domain $P_S(X, Y)$ and target domain $P_T(X, Y)$. The higher the divergence between these values the less information there is in the source domain that can be exploited to improve performance in the target domain. In the extreme case if there is no similarity, it is not possible for transfer learning to improve performance.

- The size and quality of the source and target datasets. The excess empirical loss bounds of the pretrained model are shown to be dependent on the size of the target dataset in (Liu et al. 2022). Intuitively, if labeled target data is abundant enough, a model trained on this data only may perform well. In this example, transfer learning methods are more likely to impair the target learning performance. Conversely, if there is no labeled target data, as in the domain adaptation task, a bad transfer learning method would likely perform better than a random guess, which means negative transfer would not happen.

The importance of similarity between the source and target datasets and the size of the target dataset is shown when transferring between ImageNet 1k and the COCO object detection dataset (200,000 images and 1.5 million labeled object instances). Experiments in (He et al. 2019; Zoph et al. 2020) show that using weights pretrained on ImageNet 1K results in negative transfer in many experimental settings. Further results show that doing self-supervised pretraining on the target dataset only, never results in negative transfer (Zoph et al. 2020). These results are expected as the source and target datasets are quite dissimilar as they are entirely different tasks, and COCO has slightly more labeled examples than ImageNet 1K.

With deep neural networks, once the weights have been pretrained to respond to particular features in a large source dataset it will be difficult to change these weights far from their pretrained values (Neyshabur et al. 2020). This is particularly so if the target dataset is orders of magnitude smaller as is often the case. This premise allows transfer learning to improve performance and also allows for negative transfer. If the weights transferred are pretrained to respond to unsuitable features then this training will usually not be fully reversed during the fine-tuning phase and the model could be more likely to overfit to these inappropriate features.

Most cases of poor and negative performance for transfer learning occur when all the following elements are present:

1. The target dataset is large enough to be trained reasonably well from random initialization (He et al. 2019).
2. The target and source datasets are sufficiently dissimilar that weights pretrained on the source dataset are unlikely to be a good fit for the target dataset (He et al. 2019).
3. The transfer learning solutions used do not allow the weights to change far enough from their pretrained values. This point is further discussed in Sect. 7 on best practice.

In image classification models, features learned through lower layers are more general, and those learned in higher layers are more task specific (Yosinski et al. 2014; Lee et al. 2023). It is likely that if less layers are transferred negative transfer should be less prevalent, with training all layers from random initialization being the extreme end of this. However, there has been limited prior work to test this.

3.6 Categorization of transfer learning

General transfer learning has traditionally been categorized in several ways see Fig. 3 for an overview. These categorizations include:

1. Label based categorization which looks at the availability of labels and divides transfer learning into the following categories:

- Inductive transfer learning - the source and target tasks are different and labels are available for at least the target dataset. Labels may or may not be available for the source domain.
- Transductive transfer learning - the source and target tasks are the same but the domains are different. Labels are available for the source dataset only. Referred to as domain adaptation.
- Unsupervised transfer learning - labels are not available for either dataset.

For a full exploration of this categorization see (Pan and Yang 2009).

2. Consistency between source and target data (Weiss et al. 2016).

- When $\mathcal{X}_S = \mathcal{X}_T$ and $y_S = y_T$ this is referred to as homogeneous transfer learning.
- When $\mathcal{X}_S \neq \mathcal{X}_T$ or $y_S \neq y_T$ this is heterogeneous transfer learning.

Traditionally, homogeneous transfer learning is said to have the same feature and input space. However, when extending this concept to DTL care must be taken as the feature space is learned from the data. It is common in DTL for computer vision to have $\mathcal{X}_S = \mathcal{X}_T$. This means usually either the input dimensions and ranges are equal or they are preprocessed to be. However usually $\mathcal{P}(\mathcal{X}_S) \neq \mathcal{P}(\mathcal{X}_T)$ and $y_S \neq y_T$ meaning the distributions of

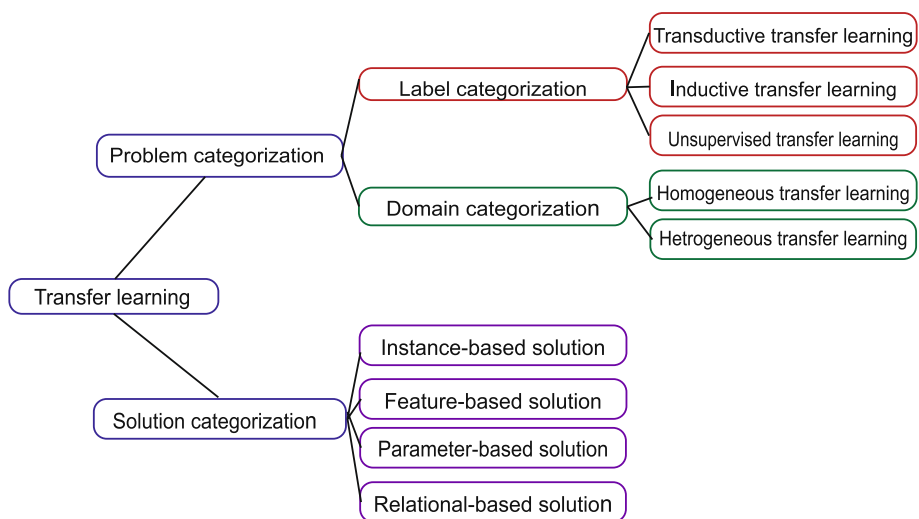


Fig. 3 General transfer learning taxonomy (Zhuang et al. 2020)

input and target values are different; therefore new features must be learned from the target dataset.

3. Solution categorization (Pan and Yang 2009; Zhuang et al. 2020) divides transfer learning into categories based on approaches to solving the task:

- Instance-based approaches - mostly based on weighting instances in the source dataset when training on the target task.
- Feature-based approaches - transform the source features to create a new feature representation for the target task, or learn a shared feature space.
- Parameter-based approaches - transfer knowledge from the source to the target dataset via the model/parameters.
- Relational-based approaches - deal with the relational domain, generally focusing on transferring the relationship via logical rules.

In DTL by definition, the features are learned and in some form the pretrained model is always used. As a result, it can be argued that all deep learning methods fall under parameter-based approaches as per (Zhuang et al. 2020). The other three approaches have been used successfully in conjunction with parameter-based approaches (Zhu et al. 2021; Ngiam et al. 2018; Mahajan et al. 2018). When considering feature-based approaches it is important to keep in mind that deep learning features are learned, in contrast to precalculated features used by traditional machine learning approaches. This means that feature-based approaches for DTL generally involve learning a shared feature space or methods that restrict or encourage the source and target feature spaces to be similar. A recent successful example is the CLIP algorithm for text to image generation (Radford et al. 2021). This algorithm encourages the text embedding space to be as close as possible to the image embedding space. It uses a contrastive loss that maximizes the dot product of the two embeddings if the caption belongs to the image and minimizes it if it belongs to a different image.

3.7 Categorization of deep transfer learning for image classification

As shown above, the traditional categorizations of transfer learning are not always well suited when applied to DTL, particularly in relation to image classification. We therefore propose a new categorization.

From Sect. 3.1 we have $f_{\mathcal{F}}^*$ the optimal function in $\mathcal{F}$ the family of functions defined by our model architecture. By Definition 1 we have $\mathcal{D}_S \neq \mathcal{D}_T$, or $T_S \neq T_T$. This implies that the optimal function for the source task is not equal to the optimal function for the target task $f_{S\mathcal{F}}^* \neq f_{T\mathcal{F}}^*$ and means that the optimal weights of the model must be learned based on the target dataset. Usually in the transfer learning setting, the target dataset is orders of magnitude smaller than the source dataset. From equation 1 we are trying to minimize the sum of the approximation error ε_{app} and the estimation error ε_{est} . If the relationship between the input data and labels is complex a model with a large number of parameters (usually in the millions or billions) is needed to ensure that the optimal function in $\mathcal{F}$ is as close as possible to the optimal function in general and reduce ε_{app} . The larger the model the more capacity it has to overfit and the larger ε_{est} can become. This means deep learning for complex tasks like image classification involves selecting a model that is large enough to learn the

relationships between the input and label data and regularizing it to ensure the effect of ε_{est} is reduced as far as possible. Pretraining a large deep learning model is a form of regularization as it creates a flatter optimization landscape with smaller gradients (Liu et al. 2019). This allows for small updates to the weights based on the target dataset while discouraging large updates. The effect can be magnified by adding a regularization term that explicitly decays the weights towards their pretrained values (Xuhong et al. 2018).

Consequently, with DTLIC in particular, we end up with a balance between:

1. Allowing the pretrained weights to move towards the weights of the optimal function of the target task $f_{T\mathcal{F}}^*$.
2. Preventing the new weights from overfitting the empirical loss approximation $R(f_n)$.

Our new categorization of DTLIC, as shown in Fig. 4, both truncates and expands on the traditional transfer learning categories.

3.7.1 Domain categorization

As we are concerned with the final task of image classification it is assumed that we have labeled examples in:

- The source domain but not the target domain - domain adaptation
- The target domain but not the source domain - self-supervised pretraining
- Both domains

Given this assumption, we removed the unsupervised transfer learning setting from the label based categorization.

We define two new categories.

Similarity: In DTL the features are learned from the data. The categorization of homogeneous versus heterogeneous thus becomes a solution categorization of whether to continue training the pretrained weights or not during the fine-tuning process so we remove it. We argue that the optimal decision in the solution space still depends on the consistency between

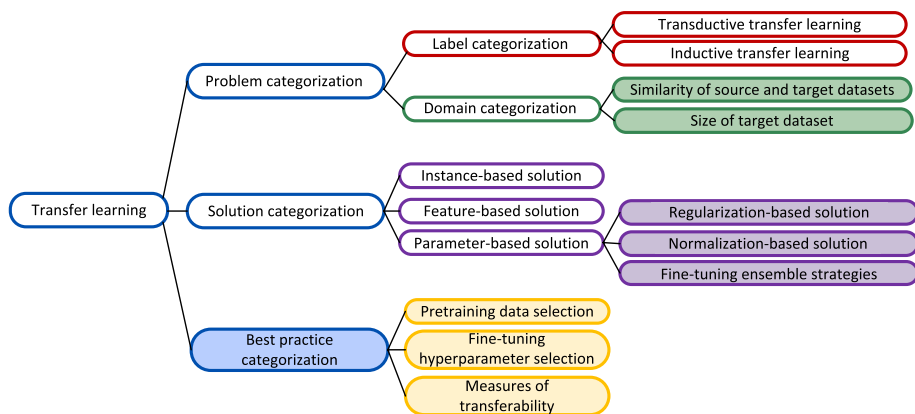


Fig. 4 Deep learning for image classification taxonomy. Shaded cells are additions to existing transfer learning and deep transfer learning taxonomies

source and target data - including both the input data and task. The more f_{TF}^* differs from f_{SF}^* the less likely that f_{SF}^* will be an optimal substitute for f_{TF}^* . To represent this in our taxonomy, we replace homogeneous versus heterogeneous with similar versus dissimilar.

Size: The balance described above of allowing the weights to move but not overfit points to another important categorization in DTL for image classification - the size of the dataset. The larger the model compared to the size of the target dataset the more likely it is to overfit if the weights are allowed to move further from their pretrained values. To represent this element we added large versus small dataset under domain categorization.

The two new categorizations above are not discrete so we also present Fig. 5 representing them on a continuous scale. The insights shown in this diagram are discussed further in Sect. 7.

3.7.2 Solution categorization

In the solution categorization space, we remove relational-based approaches. On the rare occasions when relational-based tasks are seen in the image classification task domain, their solutions generally involve the transfer of learned parameters, for example, graph neural network approaches, not logical rules.

Some previous DTL taxonomies (Tan et al. 2018; Iman et al. 2023; Fuchao et al. 2022) include a generative adversarial training (GAT) categorization. This refers to a wide variety of techniques that use adversarial training to improve results in transfer learning tasks

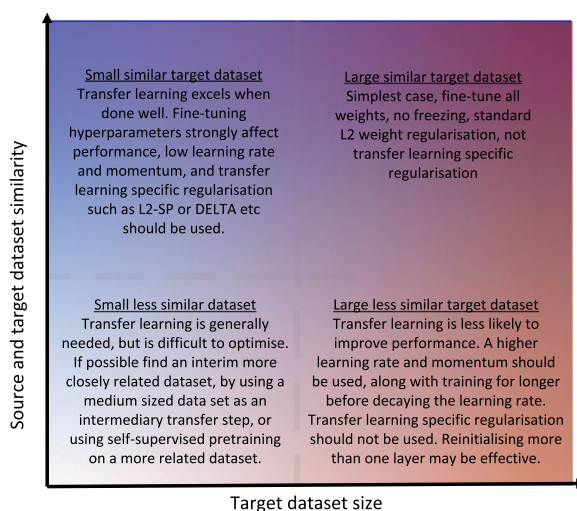


Fig. 5 Transfer learning taxonomy. Blue is a measure of source and target dataset similarity, where the most similar would be transferring from one set of ImageNet classes to another such as in (Yosinski et al. 2014; Plested and Gedeon 2019) or self-supervised pretraining on the target dataset (self-training). Red measures the target dataset size, with small being around 10,000 labeled examples and large being a million plus labeled examples. The lines between the quadrants are gray and dotted and the colors gradually merge to indicate that there is not a clear cutoff point from one quadrant to the next, rather as the target dataset increases in size between the range of around 50,000 to 200,000 more strategies for large target datasets should be employed. The number of trainable parameters in the model architecture should be considered when deciding whether to use strategies for small or large datasets. Larger models generally require more training data to avoid overfitting the target task, as highlighted in Fig. 1

(Tzeng et al. 2017; Shrivastava et al. 2017; Liu and Tuzel 2016; Attia et al. 2020). We argue that the significant uses of GAT in transfer learning are better described by alternative solution categories or by areas only peripherally related to DTLIC.

The use of GAT for transfer learning often falls into other solution categories depending on its use. For example, aligning the source and target feature spaces (Tzeng et al. 2017; Liu and Tuzel 2016; Gamrian and Goldberg 2019) and better matching of synthetic data with real data (Shrivastava et al. 2017; Taigman et al. 2017) are feature-based approaches and are discussed in Sect. 5

GATs can also be used in ways peripheral to transfer learning, such as for data augmentation. For example, generating new conditional samples from the source domain that are as similar as possible to the target domain (Attia et al. 2020), or improving the realism of synthetic samples generated another way (Shrivastava et al. 2017) are examples of data augmentation. While they can be effective when applied to transfer learning, these techniques are beyond the scope of this survey. See (Biswas et al. 2023) for a detailed overview of the use of data augmentation using GATs for image classification.

As many DTLIC approaches can be classed as parameter-based approaches, we further categorize them as follows:

Regularization based approaches: assume that the optimal solution is not too far from the pretrained solution and so the model weights or features should be prevented from moving too far during fine-tuning.

Normalization based approaches: seek to better align fine-tuning in the target domain with the source domain by making adjustments to the standard batch normalization or other forms of normalization that are used between layers in modern deep learning models.

Fine-tuning ensemble strategies combine multiple fine-tuning or weight freezing strategies to optimize the final fine-tuned model.

Note that some previous DTL surveys have included solution categories for fine-tuning and freezing weights (Iman et al. 2023; Fuchao et al. 2022). These are fine-tuning hyperparameter decisions and fall under our new categorization of best practice introduced below.

3.7.3 Best practice categorization

Finally, we add a new root categorization of best practice. To this, we add the following subcategories:

- Pretraining data selection
- Fine-tuning hyperparameter selection
- Measures of transferability.

3.8 Areas related to deep transfer learning for image classification

Several different problem domains are either closely related to deep transfer learning or regularly used as part of solutions.

3.8.1 Domain adaptation

Domain adaptation (DA) could be considered the most closely related problem domain. In DA the task T remains the same, but the Domain D differs between source and target task. According to the definition 1 by Pan and Yang (Pan and Yang 2009) DA falls under general transfer learning as transductive transfer learning as described in Sect. 3. Others define it as a separate area (Goodfellow et al. 2016). Either way DA is not a focus of this review as it is rare in image classification that the source and target domains have the same class labels. This situation is more common in natural language processing where for example, a sentiment analysis model could be transferred between two slightly different product review domains (Wang and Deng 2018; Zhang et al. 2019). The few datasets that are used in DA for image classification are carefully curated to have identical object classes in different domains (Gong et al. 2012; Saenko et al. 2010; Tommasi et al. 2014).

Despite there being few examples of DA problems in image classification, there are techniques that have been developed for these tasks that could be more broadly useful to deep transfer learning for image classification. Wang and Deng state that “When the labeled samples from the target domain are available in supervised DA, soft label and metric learning are always effective” (Wang and Deng 2018). Using an interim fine-tuning step which incorporates metric learning on the target dataset before switching to training on the final classification task has been shown to improve performance on small fine-grained target datasets (Ridnik et al. 2021).

3.8.2 Concept drift and multitask learning

The concept drift problem is related to DA in that it deals with adapting models to distributions that change over time. Multitask learning is another related problem domain where the focus is on learning one model that can perform well at multiple tasks with different but related target datasets. While concept drift or multitask learning can be seen as types of transfer learning they have a different aim to the usual transfer learning scenario. That aim is to perform well on more than one task, rather than only the final target task. The major practical difference between general transfer learning and concept drift or multitask learning is that usually catastrophic forgetting is not a consideration in the former.

3.8.3 K-shot or few shot learning

The K-shot (also referred to as few shot) learning problem domain is specifically focused on learning from very few or even zero labeled examples. This can be considered a specialized sub area of transfer learning and DA. The focus of new work in this domain is generally on improving metrics for defining and learning the best embedding spaces and comparisons to new examples, plus new data augmentation techniques. Transfer learning is implicit in most of the solutions, as weights are pretrained on a related task (Wang et al. 2020; Kaya and Bilge 2019).

While there is a lot of similarity, K-shot learning methods focus only on very small target datasets. This means they do not consider how differences in the size of the target dataset affects the optimal transfer learning methods and hyperparameters.

3.8.4 Self-supervised learning

In machine learning, unsupervised learning refers to solutions used when there are no labeled training examples. In deep learning, we define self-supervised learning as creating target labels from a dataset without training labels. This allows the use of supervised learning methods. An aspect of the data itself is used as the training signal. The most general example is autoencoders which consist of an encoder and decoder with the target being the same as the input. The middle encoding is regularized to encourage a useful semantic encoding. Other examples of task specific self-supervised learning include predicting the next frame of a video (Srivastava et al. 2015), predicting the next word in a sentence (Brown et al. 2020), image inpainting or upsampling (Pathak et al. 2016; Ledig et al. 2017), generative models (Radford et al. 2015; Goodfellow et al. 2014) and many more. For a detailed treatment of self-supervised methods for learning image representations see (Jing and Tian 2020).

Self-supervised learning relates closely to transfer learning as it is often used for pretraining when there is limited, closely related labeled data. The intent is that useful relationships in the data can be learned with self-supervised pretraining before fine-tuning on the labeled data.

3.9 Datasets commonly used in transfer learning for image classification

We anticipate that providing a comprehensive description of common source and target datasets will assist the reader in finding the best datasets for their needs.

3.9.1 General

General image classification datasets contain a variety of classes with a mixture of super-ordinate and subordinate classes from many different categories in WordNet (Miller 1995). ImageNet is a canonical example of a general image classification dataset.

3.9.2 Fine-grained

Fine-grained image classification datasets include classes with subtle differences between them for example, makes and models of cars (Krause et al. 2013) and aircraft (Maji et al. 2013), and breeds of cats and dogs (Parkhi et al. 2012).

3.9.3 Scenes

Scene datasets contain examples of different indoor and/or outdoor scene settings.

Commonly used source datasets are described in Table 4 and target datasets in Table 5.

3.9.4 The transferability estimation benchmark

Bolya et al. (2021) A benchmark for testing transferability measures. It consists of six target datasets, eight source datasets and four architectures. The target datasets include Caltech101, CIFAR10, Birds, CUB, Oxford Pets and Stanford Dogs. The source datasets include the six

Table 4 Commonly used source datasets

Dataset	Size/s	Category	Number of classes
ImageNet (Deng et al. 2009)	1.3 million+	General	1000–21,000
JFT (Hinton et al. 2015)	300 million	General	18,291
Instagram Hashtag (Mahajan et al. 2018)	3.5 billion	General	1500+
Places365 (Zhou et al. 2017)	1.8–8 million	Scenes	365
Inaturalist (Van Horn et al. 2018)	859,000	Plants and animals	5000+

Table 5 Commonly used target datasets

Dataset	Size/s	Category	Number of classes
CIFAR Krizhevsky (2009)	50,000	General	10 and 100
PASCAL VOC 2007 (Everingham et al. 2007)	9063	General	20
Caltech (Fei-Fei et al. 2004; Griffin et al. 2007)	9146	General	101 and 250
Food-101 (Food) (Bossard et al. 2014)	101,000	Food objects	500
Birdsnap (Birds) (Berg et al. 2014)	49,829	Species of birds	500
Stanford Cars (Cars) (Krause et al. 2013)	16,185	Models of cars	196
FGVC Aircraft (Aircraft) (Maji et al. 2013)	10,000	Models of aircraft	100
Oxford-IIIT Pets (Pets) (Parkhi et al. 2012)	7049	Breeds of cats and dogs	37
Oxford 102 Flowers (Flowers) (Nilsback and Zisserman 2008)	8189	Types of flowers	102
Caltech-usc Birds (CUB) (Wah et al. 2011)	11,788	Species of birds	200
Stanford Dogs (Dogs) (Khosla et al. 2011)	20,580	Breeds of dogs	120
SUN397 SUN (Xiao et al. 2010)	108,754	Scenes	397
MIT 67 indoor scenes (Quattoni et al. 2009)	15,620	Indoor scenes	67
Describable textures DTD (Cimpoi et al. 2014)	3760	Texture adjectives	47
Daimler pedestrian (Munder and Gavrilu 2006)	23,520	Pedestrians	2
German road signs (Stallkamp et al. 2012)	39,209	Road signs	43
Omniglot (Lake et al. 2015)	1.2 million	Handwritten characters	1623
SVHN digits in the wild (Netzer et al. 2011)	73,257	Street view digits	10
UCF101 dynamic images (Soomro et al. 2012)	9537	Actions	101

target datasets plus ImageNet 1k and VOC2007. The model architectures include ResNet 50 and 18 He et al. (2016), GoogLeNet Szegedy et al. (2015) and AlexNet Krizhevsky et al. (2012).

3.9.5 Visual PETL benchmark (Xin et al. 2024)

A benchmark for testing vision based parameter-efficient transfer learning (PETL) methods. It consists of 30 diverse, challenging, and comprehensive datasets from image recognition, video action recognition, and dense prediction tasks.

3.9.6 Visual decathlon challenge (decathlon) (Rebuffi et al. 2017)

A challenge designed to simultaneously test performance on 10 image classification tasks being: ImageNet, CIFAR-100, Aircraft, Daimler pedestrian, Describable textures, German traffic signs, Omniglot, SVHN, UCF101, VGG-Flowers. All images resized to low resolution with a shorter side of 72 pixels.

3.10 Real-world applications of deep transfer learning for image classification

Most real world image classification data does not have the millions or even billions of independent labeled examples needed to train a complex deep learning model well from random initialization. For this reason transfer learning is an essential part of a broad range of applications. Some examples of where transfer learning is applied are highlighted below, but there are many more:

1. *Medical images.* DTLIC is often applied in the medical image domain as datasets are generally small. There are two major concerns in this domain:
 - (a) Medical images are very different in style to image domains where data is abundant, so large, closely related source datasets are usually unavailable.
 - (b) Even within the same task in the medical image domain, there are differences stemming from the imaging machine and types of patient populations. These differences can mean that models do not generalize well to new data and must be fine-tuned on a limited amount of data that more closely matches the target data.

See (Kora et al. 2022; Kumar et al. 2024; Juodelyte et al. 2024) for a detailed overview of transfer learning and transferability in the medical image domain.

2. *Robotics.* It is more common for image classification to be part of a pipeline or backbone of a model in vision-based robotics rather than the end task. For instance, classification of objects and scene recognition can be important for navigation tasks. Action tasks can also be encoded as image classification, by encoding directions as discrete classes. Transfer learning is common in this domain, as the differences between the source and target data can come from three elements: changes to robots, tasks, or environments. If any of these items change significantly, it is likely that the model will not generalize to the new data and will need to be fine-tuned to achieve optimal performance. When the model does need fine-tuning, the aim is to generally use limited data as real world robots can be expensive and time consuming to run. While simulation environments are usually cheap and efficient for creating large amounts of data, they are generally not the final target environment, so they are far more likely to be used as the source task rather than the target task. See (Jaquier et al. 2025) for a review of transfer learning for robotics.
3. *Environmental modeling.* When we consider the format of image data as input into a deep learning model we can broaden our understanding of what an image is to include most two or three dimensional spatially related data. Viewed in this way, there are many tasks in environmental modeling that can be represented well as image classification

tasks. The tasks in environmental modeling are often complex. In addition, there is often a large amount of unlabeled data available, for example, satellite imagery, but an extremely limited amount of data available for the exact target prediction task. An example of this scenario is pyrocumulonimbus (PyroCb) prediction, where satellite images and other spatial domain information are used to predict whether a fire will become a PyroCb - i.e. create its own weather system, making its spread more erratic (Tazi et al. 2022). There are huge amounts of satellite images showing fires, but these specific events are very rare and complex to predict. For more details on this area (Yu et al. 2025) discusses transfer learning techniques throughout the review.

4. *Agriculture*. Image classification tasks are seen throughout the agriculture domain, including classifying types of crops, predicting yields, finding or classifying pests, and many more. In many of these tasks there is large amounts of unlabeled data available, but extremely limited data for the exact prediction task, for example, labeled images of a particular type of crop pest. For a more detailed discussion of deep transfer learning for image classification in this area see (Ismail et al. 2025; Olorunfemi et al. 2024).

4 Instance based approaches

In the realm of DTLIC instance based approaches look at:

- What source data should be used for pretraining and fine-tuning
- Whether particular source data instances or classes should be weighted more heavily during training.

In general, increasing the size of the source dataset increases the performance on the target dataset. This occurs even when the target dataset is large such as ImageNet 1K (1K classes, 1.3M training images) and ImageNet 5k (5K classes, 6.6M training images) or 9k (9K classes 10.5M training images) (Ngiam et al. 2018; Mahajan et al. 2018; Kolesnikov et al. 2020; Liu et al. 2022). However, pretraining on source data that, naturally or through careful curation, matches the target data more closely often performs better than pretraining on larger, more general source datasets (Ngiam et al. 2018; Mahajan et al. 2018; Xuhong et al. 2018; Sabatelli et al. 2018).

The results listed above also apply to tasks related to image classification such as object detection and semantic segmentation (Li et al. 2021; Chen et al. 2018; Gao et al. 2022; Mo et al. 2022; Zoph et al. 2020)

4.1 Self-supervised pretraining techniques

Self-supervised learning (SSL) techniques can be divided into two categories. invariant SSL (I-SSL) and equivariant SSL (E-SSL). The aim of I-SSL is to encourage the representation to be invariant to input augmentations (e.g., color jittering). Contrastive learning (CL) as the dominant invariant representation learning method, utilizes various augmentations, including random crop, horizontal flip, color jitter, and Gaussian blur. The objective is to embed semantically the same inputs obtained through augmentations close together and different inputs further apart. This is based on the concept that differences due to these augmentations

are inconsequential on a semantic level. E-SSL learns representations that are sensitive to the applied transformations. Many older SSL methods are equivariant, for example, Rot-Net (Komodakis and Gidaris 2018), which predicts the rotation angle of an image, Jigsaw (Noroozi and Favaro 2016) that arranges pieces of an image into the correct ordering, and Relative Patch Location (Doersch et al. 2015) which predicts the relative location of two patches from the same image.

Recently, invariant representations have been shown to lose augmentation-related information, for example, color information, which hinders their performance on target tasks (Lee et al. 2021; Gupta et al. 2023). This has resulted in more focus on E-SSL algorithms that learn features that are augmentation aware. Wang et al. (2024), Jaemyung et al. (2024). In many cases, algorithms that combine I-SSL with E-SSL techniques or I-SSL with an auxiliary loss term that predicts augmentations have achieved better performance than pure I-SSL or E-SSL techniques (Jaemyung et al. 2024; Lee et al. 2021; Wang et al. 2024).

As an alternative E-SSL approach, recent work has focused on creating an effective tokenizer of meaningful chunks for images, equivalent to word tokenizers that have been so successful when training large language models (Bao et al. 2021; Ramesh et al. 2021). Ramesh et al. created a codebook by training a variational autoencoder (VAE) (Kingma and Welling 2014) on the unlabeled dataset (Ramesh et al. 2021). This technique has been used successfully in many more recent self-supervised learning algorithms designed for image datasets.

Once the data and masked encoding task is properly encoded finer details of the task and training regime also have a significant impact on the final result (He et al. 2022). Considerations include:

- *Task difficulty* significantly affects the target task performance, particularly when the pretrained weights are frozen when training on the target task. For example, when the task was masked prediction, the ideal masking ratio was shown to be higher than would be intuitive at 75% (He et al. 2022).
- *Pretraining epochs*. More pretraining epochs tend to improve target task performance (He et al. 2022).

Features learned from tasks incorporating tokenization tend to show more diversity in practice and benefit target datasets requiring fine-grained semantics (Wang et al. 2024).

There is a wide range of self-supervised learning algorithms. While algorithms that encourage some form of equivariance tend to result in better performance on the target task in many cases, there are many variations with different strengths and weaknesses. For this reason, it is often better to use a combination of different self-supervised learning strategies (Huang et al. 2023). See Gui et al. (2024) for a detailed description of different supervised learning algorithms.

4.1.1 Self-training

In the absence of a much larger, closely related source dataset, self-supervised pretraining on the target dataset (self-training) before supervised fine-tuning can perform significantly better than pretraining on a less related much larger source dataset (El-Nouby et al. 2021; He et al. 2022). El-Nouby et al. (2021) have shown that self-supervised learning on smaller

target datasets can also be effective. Their self-training method was more effective than pre-training on ImageNet 1K for the COCO object detection target task (Lin et al. 2014), and comparably effective or marginally worse for a range of small image classification target tasks very different from ImageNet 1k. Further research is required to determine the limits of the self-training regime's effectiveness, specifically in terms of the target dataset's size and the extent to which it differs from any available source dataset, to establish whether self-training on the target dataset alone is the most suitable option.

4.2 Ranking of source datasets

All things being equal, more source data for pretraining is better than less, and a source dataset more similar to the target dataset is better. When the source dataset is similar enough labeled data is preferred to unlabeled data and uncured or psuedo labels are generally preferred to no labels. However, when the source dataset is less similar self-supervised learning is likely to perform better than supervised learning. Recent work has attempted to bridge this gap by allowing for multimodal class distributions by enforcing similarity in the feature space between most, not all, data points with the same class label (Feng et al. 2021). However, when the target dataset is less similar to the source dataset self-supervised pretraining is still marginally better.

The question then becomes what should be preferred when trade-offs exist between these items. This list ranks the expected performance for types of source datasets and associated techniques:

1. Supervised training on a very similar labeled dataset that is orders of magnitude larger than the target dataset. All things being equal supervised pretraining generally performs best.
2. Supervised training on a closely related uncured or psuedo labeled dataset that is orders of magnitude larger than the target dataset. If formal labels are not available, self-supervised learning is generally improved using any type of informal labels. Image tags taken from uncured databases such as Instagram and Flickr are examples of informal labels. Training with pseudo labels created by a collection of task-specific expert models can significantly improve performance on a target task compared to self-supervised training or training with uncured image tags (Salehi et al. 2023).
3. Self-supervised training on a closely related dataset with no labels that is orders of magnitude larger than the target dataset. When there is no large labeled dataset available, self-supervised pretraining on a large unlabeled source dataset will likely perform better than using an order of magnitude smaller labeled dataset or a less related dataset (Zoph et al. 2020; Azizi et al. 2021).
4. Self-training on the target dataset. Where there is no closely related source data, self-training (pretraining on the target dataset) is preferred to pretraining on a less closely related source dataset (El-Nouby et al. 2021; He et al. 2022).
5. Self-supervised pretraining on a less closely related source dataset. When the source dataset is less closely related self-supervised pretraining is likely to outperform supervised pretraining as it will not overfit the less related dataset.
6. Supervised training on a less closely related target dataset.

The above list does not include multi-step fine-tuning processes, which, if well chosen, are likely to improve performance compared to single step processes. For example, self-supervised pretraining on a closely related source dataset that is many orders of magnitude larger than the target dataset then fine-tuning on an interim dataset that is less than an order of magnitude larger would be expected to result in better performance than simple pretraining on a closely related labeled dataset that is only one order of magnitude larger (Gonthier et al. 2021; Ng et al. 2015; Azizi et al. 2021; Reed et al. 2022).

4.3 Using dataset similarity measures in transfer learning

Many studies have used dataset similarity or other measures to improve the results of transfer learning. Dataset similarity measures have been effectively used to increase performance on the target dataset by finding the most similar source dataset (Cui et al. 2018), source dataset classes (Ge and Yu 2017; Liu et al. 2022; Ngiam et al. 2018), or individual samples. Reinforcement learning methods have been used to learn to weight samples from the source dataset for effective pretraining (Zhu et al. 2020; Yoon et al. 2020). Finally, dataset similarity and transferability measures have also been used successfully to pick the most effective pretrained model for a given target task (You et al. 2021, 2022; Puigcerver et al. 2020). Transferability methods are discussed further in Sect. 7.1. The success of all these techniques for effectively using the most similar source data further showcases the importance of considering the similarity of the source and target datasets when pretraining and fine-tuning.

4.4 Large similar target datasets

All things being equal, pretraining on more source data is better, even when the target task is large, such as ImageNet 1 K (Sun et al. 2017; Mahajan et al. 2018; Dehghani et al. 2023). This is particularly so when the target task is quite different to the source task, for example transferring from an image classification task to an object detection task (Sun et al. 2017).

However, additional pretraining data is useful only if it is well correlated to the target task. (Huh et al. 2016; Mahajan et al. 2018). In some cases, adding additional unrelated training data can hinder performance.

Multi-step fine-tuning pipelines that start with pre-training on a large, less related dataset, then fine-tune on one or more smaller, more closely related datasets, before finally fine-tuning on the target task, are often more effective than single-step fine-tuning procedures (Yalniz et al. 2019; Xie et al. 2020). Incorporating pseudo labeling into these pipelines has also been shown to be beneficial (Xie et al. 2020), particularly for target tasks specifically designed to be more difficult, such as ImageNet A, B, and P (Hendrycks and Dietterich 2018; Hendrycks et al. 2021).

The size of the source dataset has become even more important recently with the introduction of vision transformers with orders of magnitude more parameters than traditional CNNs (Dosovitskiy et al. 2020; Jiahui et al. 2022). Large vision transformers, or a combination of transformers and CNNs, have shown superior performance to stand-alone CNNs in cases where the target dataset is large, for example, ImageNet1K, and the source dataset is orders of magnitude larger (Dehghani et al. 2023; Chen et al. 2023; Jiahui et al. 2022).

The introduction of self-training techniques has emphasized the importance of pretraining data being closely related to the target dataset to improve performance (El-Nouby et al. 2021; He et al. 2022). Self-training methods can be particularly effective when applied to the largest models, such as vision transformers (El-Nouby et al. 2021).

4.5 Large target datasets with less similar tasks

There are very few large image datasets that are not closely related to the image classification tasks that are usually used for pretraining. The Places365 with 1.8 million training images was used as a source task in (Mahajan et al. 2018). It was shown that with less related source and target datasets the bigger and more diverse the source training dataset the better the results on the target dataset. It has also been shown that self-supervised pretraining can be better than supervised pretraining when the source dataset is less related (Azizi et al. 2021; Feng et al. 2021).

Raghu et al. (Raghu et al. 2019) showed that when medical image datasets are large (around two hundred thousand training examples), pretraining on ImageNet 1 K results in limited improvements over random initialization. When the number of training examples was reduced to a small dataset size of 5,000, pretraining with ImageNet 1 K resulted in small improvements over random initialization.

There have been mixed results shown on whether pretraining with ImageNet 1 K and other common source datasets improves object detection performance on the COCO dataset (Lin et al. 2014) which has 328,000 training images and 1.5 million bounding boxes across 80 classes. In recent years, several larger object detection datasets have been introduced (Kuznetsova et al. 2020; Shao et al. 2019; Cai et al. 2022). Shao et al. (2019) introduced the object detection dataset Objects365 which contains 365 object categories, 638 thousand images, and 10 million bounding boxes. The authors found that for the target task COCO, pretraining on Objects365 was more effective than pretraining on ImageNet 1 K and resulted in a significantly shorter fine-tuning time. Additionally, the multi-step pretraining process of pretraining on ImageNet 1 K, then Objects365 before transferring to the COCO target dataset slightly improved results over a single pretraining step.

4.6 Smaller target datasets with similar tasks

Similarly to large target datasets, better matched pretraining data has been shown to produce better performance on the target task than more pretraining data (Ngiam et al. 2018; Puigcerver et al. 2020; Xuhong et al. 2018; Cui et al. 2018; Liu et al. 2022; Sabatelli et al. 2018; You et al. 2020).

When pretraining deep neural networks on ImageNet 1 K (Deng et al. 2009) and transferring to significantly smaller target datasets, the improvement of DTL over random initialization correlates positively with the similarity of the source dataset to the target dataset. The improvement correlates negatively with the size of the target dataset. Both correlations are seen strongly in (Kornblith et al. 2019), with the positive correlation of performance improvement using transfer learning with the similarity between source and target datasets being most obvious. The authors state that on Stanford Cars and FGVC Aircraft, the improvement was unexpectedly small. The improvement over equivalent models trained from random initialization is marginal for these two datasets, at 0.6% and 0.2%. Stanford

Cars and FGVC Aircraft both contain fine-grained makes and models of cars and aircraft respectively, whereas ImageNet 1 K contains no makes and models, just a few coarse categories for both. This means that the similarity between the source and target datasets for these two tasks is much lower than, for example, the more general CIFAR-10/100 or Oxford-IIIT Pets for which there are actually slightly more fine-grained classes in the source than the target dataset.

It is interesting to note that the improvement in using fine-tuning over fixed pretrained features for Stanford Cars and FGVC Aircraft is significantly larger, at 25.4% and 25.7%, than for any of the 10 other datasets used in the experiments. This indicates that the pretrained features are not well suited to the task. For comparison, at the other end of the similarity scale, Oxford-IIIT Pets shows one of the largest improvements from 83.2% accuracy for training from random initialization to 94.5% accuracy for fine-tuning a pretrained model. For this dataset the increase in performance using fine-tuning compared to fixed pretrained weights is marginal at 1.1%. This shows that the pretrained features are well suited to the task without fine-tuning. Recent work has demonstrated several ways to effectively predict how well a pretrained model will transfer to a particular target task. These methods are covered in Sect. 7.

For smaller target datasets, large transformer models perform well where the source dataset is large and closely related to the target dataset, for example, when transferring from ImageNet 1 K or other large general datasets to smaller general datasets such as CIFAR 10 and 100 (Dosovitskiy et al. 2020; Ridnik et al. 2023; Zhou et al. 2021). They have not achieved performance comparable to the state of the art for tasks such as for Stanford Cars and FGVC Aircraft (Ridnik et al. 2023; Liu et al. 2023) where no closely related large source datasets currently exist. When transformers do perform better they have been shown to have more robust features (Mauricio et al. 2023).

4.7 Smaller target datasets with less similar tasks

Transfer learning has been shown to be effective in many areas where the target datasets are small and less related to ImageNet 1 K and other common source datasets. However, there have also been several recent results that fit into this category where DTL has shown little or no improvement over random initialization (Zoph et al. 2020; He et al. 2019; Raghu et al. 2019). Transfer learning shows better performance on smaller target datasets that are more closely related to the source dataset than larger less related datasets in general see Sect. 7 and (Kornblith et al. 2019). Self-supervised learning methods applied to source datasets that are more closely related but unlabeled, or even self-training on the target task, often performs better than supervised learning methods applied to less closely related source datasets (Zoph et al. 2020; Azizi et al. 2021; Dai et al. 2024). However, even when the target dataset is very dissimilar to the source dataset and transfer learning brings no performance gain it can accelerate the convergence speed (He et al. 2019; Raghu et al. 2019).

Multi-step fine-tuning pipelines can be particularly effective in the case of small target datasets with limited closely related data (Gonthier et al. 2021; Azizi et al. 2021), as can self-training (El-Nouby et al. 2021).

5 Feature based approaches

As discussed in Sect. 3 the deep learning interpretation of feature based approaches differs from the traditional machine learning interpretation because the feature space is learned in the former. This means that feature based approaches for DTLIC focus on creating either a shared feature space or ensuring the source and target feature spaces are as similar as possible.

A deep learning model can be trained with a metric, that measures the distance between the source and target representations, combined with the standard cross entropy classification loss. This is to achieve the dual aims of a domain invariant model that is also a strong classifier (Tzeng et al. 2017; Long et al. 2015). A commonly used metric is the Maximum Mean Discrepancy (MMD) (Borgwardt et al. 2006) and its extensions such as conditional (MMD) (Long et al. 2015) and local (MMD) (Zhu et al. 2020). Zhu et al. (Zhu et al. 2019) successfully extend this approach to multiple representations of the same image.

Generative adversarial techniques are used in several feature based approaches, including:

1. To promote similarity between the source and target domain representations (Tzeng et al. 2017; Liu and Tuzel 2016). This approach is used most often in domain transfer to create a shared source and target domain encoding space. A deep learning model maps examples from the source and target domains to a shared feature space. An adversarial loss is used to classify whether the encoded features belong to the source or target domain. This approach performs well when no target labels are available, but it is generally less efficient than using a predefined metric or target labels when they are available.
2. To map target images into the source domain or vice versa (Taigman et al. 2017; Gamrian and Goldberg 2019). These techniques aim to overcome differences in the input distributions $P(X)$ between the source and target datasets by learning a mapping from one to the other using an adversarial loss. They have been shown to be very effective at handling known domain shifts in images, for example, by denoising noisy images from the target dataset prior to classification (Gamrian and Goldberg 2019).

6 Parameter based approaches

6.1 Parameter efficient transfer learning

Large pre-trained vision (VMs) and vision-language models (VLMs) have achieved impressive results in an extensive range of image classification target tasks. Full fine-tuning, though effective, still requires substantial computational and memory resources. This is particularly costly for the largest visual-language models that have hundreds of billions of parameters, but is also becoming a problem for the largest vision only models that have tens of billions of parameters. In addition, when the model is fine-tuned for a new target task, each task requires a complete set of parameters to be stored, which can quickly become costly for multiple tasks (Zhu et al. 2024; Xin et al. 2024).

Parameter-efficient transfer learning (PETL) methods were originally designed for adapting large language models to target tasks where more limited data is available. PETL

methods seek to achieve a balance between the number of parameters trained in the fine-tuning stage, and thus efficiency, and performance. Aside from the benefits in efficiency and performance PETL methods have recently been shown to be far less likely to allow memorization and subsequent leaking of fine-tuning training data.

In the past two years, many PETL techniques have been developed for adapting pre-trained vision only or visual language models to downstream tasks (Zhong et al. 2024; Bruce et al. 2024; Xin et al. 2024; Shen et al. 2024). These methods can be grouped into four categories of approaches:

1. Prompt based approaches augment the input data with task-specific prompts. Visual Prompt Tuning (VPT) adapts this technique to visual transformers by prepending a set of soft prompts to the input tokens of each Transformer layer and only optimizes the prompts during fine-tuning (Jia et al. 2022).
2. Adapter based approaches introduce additional trainable parameters, for example, a fully connected layer block, to the frozen pre-trained model (Lialin et al. 2023).
3. Direct selective parameter tuning (DSPT) selectively updates a subset of parameters of the pre-trained model as a trade-off between full fine-tuning and linear probing. Examples are updating just the bias terms (Zaken et al. 2021), just the layer normalisation parameters, Basu et al. (2024), or a combination of both Xie et al. (2023).
4. Efficient selective parameter tuning (ESPT) methods learn additive residuals to the original parameters, for example ΔW to update W . By injecting a low-rank constraint to the residuals, this category effectively reduces the learnable parameters Hu et al. (2022).

For a more indepth discussion of visual PETL methods see Mai et al. (2024), Han et al. (2024).

6.2 Regularization based approaches

Most regularization based techniques aim to prevent fine-tuned weights from overfitting the unreliable empirical loss approximation $R(f_n)$ 3.1 by restricting the model weights or the features produced by them. They achieve this by adding a regularization term $\lambda \cdot \Omega(\cdot)$ to the loss function:

$$\min_w L(w) = \left\{ \frac{1}{n} \sum_{i=1}^n L(z(x_i, w), y_i) + \lambda \cdot \Omega(\cdot) \right\} \quad (2)$$

with the first term $\frac{1}{n} \sum_{i=1}^n L(z(x_i, w), y_i)$ being the empirical loss and the second term being the regularization term. The tuning parameter $\lambda > 0$ balances the trade-off between the two.

The success of regularization based techniques for DTL rely heavily on the assumption that the source and target datasets are closely related. This assumption implies that the optimal weights or features for the target dataset are not far from those trained on the source dataset.

There are two major categories of regularization, weight regularization and knowledge distillation.

6.2.1 Weight regularization

Weight regularization directly restricts how much the model weights can move. L2-SP regularization $\Omega(w) = \|w - w^0\|_2^2$ is the squared L2 norm between the source weights and the current weights (Xuhong et al. 2018; Li et al. 2020). L2SP regularization has been shown to be significantly more effective than decaying towards zero when the source dataset is very similar to the target dataset, but can harm performance when the target dataset is less similar (Li et al. 2019; Wan et al. 2019; Plested et al. 2021; Chen et al. 2019). More recent work has shown that in some cases using L2-SP regularization for lower layers and L2 regularization for higher layers can improve performance compared to L2-SP or L2 regularization alone (Plested et al. 2021).

6.2.2 Knowledge distillation

Knowledge distillation, or feature based regularization, uses the distance between the feature maps output from one or more layers of the source and target networks to regularize the model:

$$\Omega(w, w_s) = \frac{1}{n} \sum_{j=1}^N \sum_{i=1}^n d(F_j(w_t, x_i), F_j(w_s, x_i)) \quad (3)$$

where $F_j(w_t, x_i)$ is the feature map output by the j th filter in the target network defined by weights w_t for input value x_i , and $d(\cdot)$ is a measure of dissimilarity between two feature maps.

DELTA (Li et al. 2021) is an example of knowledge distillation or feature map based regularization. It is based on the idea of retraining CNN channels that are not useful to the target task while not changing channels that are useful. Training on the target task is regularized by the attention weighted L2 loss between the final layer feature maps of the source and target models:

$$\Omega(w, w^0, x_i, y_i) = \sum_{j=1}^N (W_j(w^0, x_i, y_i) \cdot \|FM_j(w, x_i) - FM_j(w^0, x_i)\|_2^2) \quad (4)$$

where $FM_j(w, x_i)$ is the output from the j th filter applied to the i th input. The attention weights W_j for each filter are calculated by removing the model's filters one by one (setting its output weights to 0), and calculating the increase in loss. Filters resulting in a high increase in loss are then set with a higher weight for regularization, encouraging them to stay similar to those trained on the source task. Others that are not as useful in the target task are less regularized and can change more. DELTA is shown to be slightly better than L2SP

at improving performance on target datasets that are very similar to the source dataset, but again can decrease performance when the target dataset is less similar. (Kou et al. 2020).

Attentive feature alignment (AFA) (Xie et al. 2021) improves on DELTA by calculating attention weights in the spatial domain as well as the channel level. It adds a second term to DELTA regularization that has spatial attention weights predicted by a two layer neural network:

$$\Omega_C(w, w^0, W_a^s, x_i) = \sum_{j=1}^{Height} \sum_{k=1}^{Width} \left\| A_{j,k}^i FM_j(w, x_i) - FM_j(w^0, x_i) \right\|_2^2 \quad (5)$$

Where height and width are the height and width of each feature map and $A_{j,k}^i$ is the attention weight at that position calculated by the neural network. AFA is shown to improve on L2-SP and DELTA when the source and target datasets are very similar.

Wan et al. (Wan et al. 2019) propose a method that prevents the regularization term from moving the weights in the opposite direction of the empirical loss term in the hope that this will stop negative transfer. They show that their proposal slightly improves performance with a ResNet-18 model on four different target datasets.

Batch spectral shrinkage (BSS) (Chen et al. 2019) introduces a loss penalty applied to smaller singular values of channelwise features in each batch update during fine-tuning so that untransferable spectral components are suppressed. The results show that their method never hurts performance on the given datasets and often produces significant performance gains over L2, L2-SP and DELTA regularization for smaller target datasets. They also show that BSS can improve performance for less similar target datasets where L2-SP hinders performance. Sample-based regularization (Jeon et al. 2020) proposes regularization using the distance between feature maps of pairs of inputs in the same class, as well as weight regularization. The authors report an improvement over L2-SP, DELTA and BSS in all tests. Their results reconfirm that BSS performs better than DELTA and L2SP in most cases and in some cases DELTA and L2SP decrease performance compared to the standard L2 regularization baseline.

Given the number of methods showing only small improvements compared to previous methods, more work is needed to show comparisons across a broad range of tasks and models. In particular, most work so far has focused on how DTL regularization techniques compare when the source and target dataset are very similar. More investigation is needed to show how each of these methods perform when applied to less similar datasets.

In some cases using L2-SP regularization for lower layers and traditional L2 regularization decaying the weights towards zero for higher layers can improve performance over using either one for all layers (Plested et al. 2021). While the evidence for this is so far limited it does align with observations from (Yosinski et al. 2014) that lower layers are more transferable than higher layers. Recent work shows a higher learning rate for lower layers is needed when the distribution shift between the source and target dataset is in the form of low level image corruptions (Lee et al. 2023). We recommend further investigation into whether more regularization should be applied to lower layers compared to higher layers in this case, too.

6.3 Normalization based approaches

Several techniques attempt to better align fine-tuning in the target domain with the source domain by making adjustments to the standard batch normalization or other forms of normalization that are used between layers in modern deep learning models.

1. Sharing batch normalization hyperparameters across source and target domains has been shown to be more effective than having separate ones across many domain adaptation tasks (Wang et al. 2019; Carlucci et al. 2017). Wang et al. (Wang et al. 2019) introduce an additional batch normalization hyperparameter called domain adaptive α . This takes standard batch normalization with γ and β shared across source and target domain and scales them based on the transferability value of each channel calculated using the mean and variance statistics prior to normalization. These techniques have been shown to improve performance in unsupervised domain adaptation, however as far as we are aware they have not been applied to the general supervised transfer learning case.
2. Stochastic normalization (Kou et al. 2020) samples batch normalization based on mini-batch statistics or based on moving statistics for each filter with probability hyperparameter p . At the start of fine-tuning on the target dataset the moving statistics are initialized with those calculated during pretraining in order to act as a regularizer. This is designed to overcome problems with small batch sizes resulting in noisy batch-statistics or the collapse in training associated with using moving statistics to normalize all feature maps (Ioffe 2017; Ioffe and Szegedy 2015). The results show that their methods improve over BSS, DELTA and L2-SP for low sampling versions of three standard target datasets and improve over all but BSS for larger versions of the same datasets.
3. A more efficient training mode designed specifically for fine-tuning is proposed in (You et al. 2024). The authors designed a training mode for batch normalization that they prove is equivalent to standard eval mode, but more memory and computation efficient than the standard eval mode that is usually used. Their results are also shown empirically.

In addition to the above, there are also techniques for generalising a model trained in one domain to either a related domain or out of distribution data. These techniques apply normalization approaches to align test data statistics with the training domain (Fan et al. 2021; Li et al. 2022; Nado et al. 2020). A full discussion of these techniques is beyond the scope of this work but can be found in Huang et al. (2023).

6.4 Ensemble strategies

Ensembling of different deep learning models is a well known technique to improve inference performance. When applied to transfer learning ensembling strategies can improve fine-tuning by combining more than one strategy, for example fine-tuned and frozen layers, or different fine-tuning hyperparameters.

Guo et al. (2019) make two copies of their ResNet models pretrained on ImageNet 1K. One model is used as a fixed feature selector with the pretrained layers frozen and the other model is fine-tuned. They reinitialize the final classification layer in both. A policy net trained with reinforcement learning is then used to create a mask to combine layers from

each model together in a unique way for each target example. They report state of the art results on the decathlon 10 task dataset. Their results are less competitive on other more standard datasets like Stanford Cars and Flowers, however, they do show improvements over simple fine-tuning. A similar technique with a genetic algorithm used to select which layers to freeze or fine-tune is shown to be more effective than a simple fine-tuning strategy in Nagae et al. (2022). Nagae et al. (2022) used normalized optimal transport distance between the features of the source and target dataset for each layer to decide which layers to freeze or fine-tune. This technique showed improved accuracy on the target dataset when tested on small greyscale datasets. More testing is needed to confirm this result for a broader range of datasets Nagae et al. (2022).

MultiTune (Wang et al. 2020) simplifies SpotTune by removing the policy network and concatenating the features from each model prior to the final classification layer. It also improves on SpotTune and Nagae et al. (2022) by using two different non-binary fine-tuning (Plested et al. 2021) hyperparameter settings rather than one fine-tuned and one frozen model. The results show that MultiTune improves or equals accuracy compared to SpotTune in most cases tested with significantly less training time. AMF: Adaptable Weight Fusion (Shen et al. 2022) further improves on MultiTune by adding a policy network that learns the optimal combination of the final features of each model, rather than simply concatenating them and results in further improved performance. Antonio et al. achieve similar results using a variation of adaptable weight fusion that learns a weighting of two models trained on different subsets of the source data (bagging) Antonio et al. (2025).

Ensembling can also be effective in pretraining, particularly when many similar but small source datasets are available. In Du et al. (2024), similarity between the target and potential source datasets was calculated using cosine similarity between convolutional kernels pretrained on a large source dataset. Higher accuracy was achieved when doing ensemble pretraining using all potential source datasets over a preset threshold compared to using only one source dataset.

6.4.1 Ensemble models

In transfer learning it is usual that different models initialized with the same pretrained weights will end up in the same weight basin (Neyshabur et al. 2020). Ensembles of models with different pretrained weights (global ensemble) significantly outperform those with the same weight initialization (local ensemble), particularly when the target task is less similar to the source task (Sadrtadinov et al. 2023). The weights from local ensemble models can be averaged to create a model soup (Rame et al. 2022; Wortsman et al. 2022). While model soups have been shown to have slightly inferior performance to local ensemble models on in-distribution data, they are more robust to out-of-distribution data and have significantly lower inference compute costs (Rame et al. 2022; Wortsman et al. 2022). Recent work has created semi-local model soups by averaging weights created from the same pretrained models with diverse fine-tuning hyperparameters. With a good combination of hyperparameters these semi-local model soups can partially bridge the gap between local ensembles/model soups and global ensembles, balancing efficiency and prediction performance (Sadrtadinov et al. 2023).

Supervised pretraining tends to result in better performance on downstream target tasks than self-supervised learning, all things (size and similarity of the source and target datasets)

being equal. However, there is some evidence that this may not hold with ensemble models (Sadrtadinov et al. 2023). More work is needed to explore this finding. Future work should also consider the similarity between the source and target datasets and the size of the target dataset when deciding on a model ensemble or model soup scheme.

7 Best practice categorization

There are many different questions that need to be examined when deciding on how to perform transfer learning for a particular source and target task. They can be broadly categorized by the following questions:

1. *What* type of solutions should be considered?
2. *How* should the potential solutions and training data be used?
3. *Which* hyperparameters should be used?
4. *Are* there ways to decide whether transfer learning will be successful and how to do it?

The distinction between target dataset sizes discussed in Sect. 3.7.1 is useful here as it has been shown that small target datasets are much more sensitive to changes in transfer learning settings (Plested and Gedeon 2019; Liu et al. 2022). It has also been shown that standard transfer learning protocols do not perform as well when transferring to a less related target task (He et al. 2019; Kornblith et al. 2019; Plested et al. 2021), with negative transfer being an extreme example of this (Wang et al. 2019, 2019), and that the similarity between datasets should be considered when deciding on hyperparameters (Li et al. 2019; Plested et al. 2021). These distinctions go some way toward explaining conflicting performance of DTL methods in recent years (He et al. 2019; Li et al. 2019; Wan et al. 2019; Zoph et al. 2020).

7.1 Measures of transferability

We start by answering the final question above - are there ways to decide whether transfer learning will be successful and how to do it?

Although empirical transferability can be the gold standard of describing how easy it is to transfer the knowledge learned from a source task to a target task, it is computationally expensive to obtain. A transferability metric is a function of the source and target data that approximates the empirical transferability. It is, therefore, imperative to find efficient transferability metrics that can accurately estimate empirical transferability (Tan et al. 2024).

Measures of transferability can show us how well the current model is fitting the new dataset and how far the weights need to be moved to find an optimal solution. This can help us to:

- Determine how well transferring a specific pretrained model to a new target dataset will perform
- Determine whether negative transfer is likely
- Potentially identify the best transfer learning protocols to use.

Measures of transferability can be divided into Source-Free Model Transferability Estimation (SF-MTE), which assumes no access to the source dataset, and Source-Dependent Model Transferability Estimation (SD-MTE), which assumes access to the source dataset (Ding et al. 2024).

SF-MTE methods include static methods and dynamic methods. Static methods calculate scores directly based on statistical information, such as features and logits computed on the target dataset using candidate source models. Dynamic methods compute scores using mapping functions or learning frameworks to map the original static information into a different space. Some well known static SF-MTE methods include H-Score (Bao et al. 2019), LEEP (Nguyen et al. 2020), negative conditional entropy (NCE) (Tran et al. 2019) and FERM (Fu et al. 2022).

1. Feature structure-based methods primarily focus on the intra-class and inter-class separation of features. Commonly used methods include H-Score (Bao et al. 2019).
2. Bayesian Statistic methods are based on estimating the joint distribution of source and target datasets. The most well known model family under these methods is LEEP (Nguyen et al. 2020).
3. Information theory based methods measure incorporate entropy and mutual information (Tran et al. 2019).

Dynamic SF-MTE methods usually aim to approximate changes in the model during fine-tuning using an efficient transformation. Methods in this area include LogME (You et al. 2021, 2022) and KITE (Guo et al. 2024).

SD-MTE can also be further divided into static and dynamic methods. Since the source data is available, dataset similarity methods can be incorporated. Static SD-MTE methods include many variations of the application of optimal transport distance (OT), also known as Wasserstein distance (W-distance), including OTCE, JC-NCE, F-OTCE, and JC-OTCE (Tan et al. 2021, 2024).

Dynamic SD-MTE also involves a transformation that maps the original information into a different space. In this case using dynamic features such as gradients or learning frameworks.

The principle gradient expectation (PGE) of the source and target dataset is calculated in Qi et al. (2022) and their similarity is compared to approximate the transferability.

There have been limited studies on the robustness of transferability estimators. An initial study found that minor variations in experimental setups can result in different conclusions regarding the performance of one transferability metric over another (Agostinelli et al. 2022). Another limited study found that there are no transferability metrics capable of consistently estimating the performance on the target datasets in the medical domain (Chaves et al. 2023).

It should be noted that measures of similarity introduced in Sect. 3.3, can also be used as a proxy for transferability, as the more similar a source and target dataset are, the more effective transferring a model pretrained on the source dataset to the target dataset will be.

7.2 General best practice

Transfer learning protocols have been heavily influenced by early work performed by Yosinski et al. (Yosinski et al. 2014). They found that when the source and target datasets are extremely similar and the target dataset is large (over 500,000 training examples), transferring more layers is better, and fine-tuning results in better performance than freezing layers. It is important to note that their results were only shown to hold for large and closely related target datasets and their specific fine-tuning hyperparameters. In fact, they demonstrated that higher layers are less transferable when the source and target datasets are less related. More recent work has found that transferring less layers may result in better performance when the source and target datasets are less similar (Plested and Gedeon 2019; Plested et al. 2021), and the optimal learning rate depends on the size and similarity of the target dataset compared to the source dataset (Plested et al. 2021; Lee et al. 2023).

There are some general guidelines for optimal transfer learning that do not depend on the size or similarity of the source and target datasets. These include:

1. *Fine-tuning*: tends to work better than freezing weights in almost all cases (Yosinski et al. 2014; Plested and Gedeon 2019; Huang et al. 2023). To find the optimal learning rate for fine-tuning the similarity between the source and target datasets and tasks should be considered. The layers that are likely to be most useful to the target task should also be considered (Plested et al. 2021). For example, Lee et al. found that a higher learning rate for lower layers performs better when the distribution shift is in the form of low level image corruptions (Lee et al. 2023). Recent work has shown that gradually freezing layers with low gradient norms throughout training can result in better performance than standard fine-tuning (Davila et al. 2023). More work should be done to confirm this result on a wider range of datasets.
2. *More versus closely related source data*. More source and target training data is better in general. (Ngiam et al. 2018; Mahajan et al. 2018; Kolesnikov et al. 2020). However, a smaller, closely related dataset is better than a larger, less similar dataset (Ngiam et al. 2018; Mahajan et al. 2018). Pretraining with noisy, uncurated labels or pseudo labels can improve final performance over purely self-supervised training. When a large, closely related source dataset is not available, self-training can result in significantly better performance on the target task compared to pretraining on a less related source dataset (El-Nouby et al. 2021; He et al. 2022). See Sect. 4.2 for a full ranking of source datasets and techniques.
3. *Model selection*. Model architectures that perform better on ImageNet were found to perform better on a range of target datasets in (Kornblith et al. 2019). For models trained with a wide range of self-supervised pretraining methods on a large source dataset, few shot fine-tuning performance on the source dataset has been shown to correlate closely with performance on a range of diverse downstream target tasks.
4. *Number of layers pretrained*. The number of layers to reinitialize from random weights is strongly related to the optimal fine-tuning learning rate. It is likely that a larger learning rate is optimal when more layers are pretrained and a lower learning rate when less layers are pretrained (Plested and Gedeon 2019; Plested et al. 2021). For this reason it is important to tune both the number of layers pretrained and learning rate together. Given that a higher fine-tuning learning rate on lower layers than higher layers has been shown

- to be optimal when the difference between the source and target dataset are low level image distortions more work should be done to see whether reinitializing lower layers instead of upper layers could be optimal for these types of distribution shifts (Lee et al. 2023).
5. *Regularization*. L2-SP or other more recent transfer learning specific regularization techniques like DELTA, BSS, stochastic normalization, etc. improve performance when the source and target dataset are closely related, but often hinder it when they are less related (Xuhong et al. 2018; Li et al. 2021; Wan et al. 2019; Plested et al. 2021). These regularization techniques are discussed in more detail in Sect. 6.2.
 6. *Measures of transferability*: can predict the performance of a pretrained model on a particular target task (Bao et al. 2019; Tran et al. 2019; Nguyen et al. 2020; You et al. 2022; Fu et al. 2022). The performance of a simple classification model with frozen weights can give insight into transfer learning best practice for the target task (Plested et al. 2021). More work is needed to determine if transferability measures in general can help determine optimal transfer learning hyperparameters across a wide range of source and target datasets.

7.3 Best practice for large similar target datasets

Most techniques will work well in this case, so less time needs to be spent finding the best technique for the given task (Yosinski et al. 2014; Kolesnikov et al. 2020). Transferring all layers is likely to be optimal (Plested and Gedeon 2019; Plested et al. 2021). A lower learning rate and momentum are better for fine-tuning when the source and target datasets are similar (Li et al. 2019; Plested et al. 2021; Dai et al. 2024). However, this may not be necessary when the target dataset is very large and overfitting a poor empirical loss minimizer is less of a concern. If a higher learning rate is used it should be decayed quickly so as not to change the pretrained weights too much. All weights should be fine-tuned not frozen (Yosinski et al. 2014; Plested and Gedeon 2019; Huang et al. 2023). L2-SP regularization or other more recent techniques like DELTA, BSS, stochastic normalization etc. are likely to work reasonably well (Li et al. 2019; Wan et al. 2019; Plested et al. 2021; Chen et al. 2019). However, a very large target dataset may need less regularization in general.

7.4 Best practice for large, less similar target datasets

In this scenario, negative transfer is a concern as it is much more difficult to improve performance using transfer learning. Care must be taken when selecting transfer learning protocols, and self-training on only the target dataset should be considered.

If transferring straight from a less related source dataset a higher learning rate, momentum and training for longer before decaying the learning rate should be used when fine-tuning (Li et al. 2019; Plested et al. 2021). This is to attempt to move weights out of the flat basin of the loss landscape that is created when using pretrained weights, and further away from their sub-optimal pretrained values (Neyshabur et al. 2020; Liu et al. 2019).

Recent regularization techniques like L2-SP, DELTA, BSS, stochastic normalization, etc. should not be used in this case as weights should be allowed to move freely away from their pretrained values based on the larger target dataset. Pretraining fewer layers of weights could also be attempted in order to take advantage of the more general lower layers, while

allowing the more task specific higher layers to train from random initialization (Yosinski et al. 2014; Plested et al. 2021). Alternatively fine-tuning higher layers at a higher learning rate than lower layers could take advantage of the fact that lower layers are likely to be more general and need changing less than the task specific higher layers.

Finally, a global ensemble created from models fine-tuned from different pretrained starting points should be considered (Sadrtadinov et al. 2023).

7.5 Smaller target dataset considerations overview

It is often not possible to train deep neural networks on small datasets from random initialization with supervised learning (Kora et al. 2022; Mormont et al. 2018; Kraus et al. 2017; Heker et al. 2019; El-Nouby et al. 2021; He et al. 2022). This means that transfer learning becomes more important as the size of the target dataset is reduced. It has also been shown that the transfer learning protocols, including the number of layers to transfer and fine-tuning hyperparameters have a much greater impact on performance as the size of the target dataset is reduced (Plested and Gedeon 2019; Plested et al. 2021). As the size of the target dataset decreases, two competing factors affect transfer learning performance:

1. The empirical loss estimate becomes less reliable, as described in Sect. 3.1 making overfitting idiosyncrasies in the target dataset more likely.
2. The pretrained weights implicitly regularize the fine-tuned model more strongly and the final weights generally do not move as far from their pretrained values (Neyshabur et al. 2020; Liu et al. 2019; Raghu et al. 2019).

The outcome of the first point is an increasing need to use transfer learning and other methods to reduce overfitting. The implicit regularization noted in the second point can have a positive impact in reducing overfitting the empirical loss as per the first point. It can also have a negative impact (negative transfer) if the weights transferred from the source dataset are ineffective for the target dataset. When the weights, and thus the features produced, are restricted to being far from optimal the negative effect on performance can be compounded by the first point as inappropriate features can be used to overfit the new task (Plested and Gedeon 2019).

Perhaps counterintuitively, transferring less than all available layers is often optimal when the target dataset is small, even when the source and target datasets are extremely similar. This is because it is more difficult to change large pretrained weights based on a small target dataset and so overfitting inappropriate features becomes more likely (Plested and Gedeon 2019; Plested et al. 2021).

While generally accepted practice is to train higher layers at higher learning rates a higher fine-tuning learning rate on lower layers than higher layers has been shown to be optimal when the difference between the source and target dataset are low level image distortions (Lee et al. 2023).

In general, the learning rate should be decayed quickly with smaller target datasets (Plested and Gedeon 2019; Kolesnikov et al. 2020). However, when the target data set is small it must be taken into account that the number of weight updates per epoch will be low and the number of updates should be reduced, not necessarily the number of epochs.

7.6 Best practice for small similar target datasets

This scenario is where transfer learning really excels, although more care needs to be taken to use optimal hyperparameters when the dataset is small. The experiments from (Yosinski et al. 2014) using large similar target datasets were repeated in (Plested and Gedeon 2019), but included a range of smaller target dataset sizes. These experiments demonstrated that findings for large similar target datasets do not carry over to smaller target datasets. A significant improvement in performance was shown using optimal transfer learning protocols for small target datasets compared to more commonly used protocols from (Yosinski et al. 2014). This improvement increased considerably as the size of the target dataset decreased.

For small similar target datasets the optimal learning rate and momentum will likely be low as the weights will not need to be moved far from their pretrained values and if they are, overfitting is more likely with a small dataset (Plested and Gedeon 2019; Kolesnikov et al. 2020). The learning rate should be decayed quickly for the same reason.

Recent regularization techniques like L2-SP, DELTA, BSS, stochastic normalization, etc. are likely to improve performance significantly in this case (Li et al. 2019; Wan et al. 2019; Plested et al. 2021; Chen et al. 2019).

7.7 Best practice for small less similar target datasets

In this case transfer learning can be very useful if done well, but can also lead to poor results. As highlighted in Sect. 6.2 there are many methods designed to improve performance in transfer learning that perform well when the target dataset is small and similar, but poorly when the target dataset is less similar (Nguyen et al. 2020; Xuhong et al. 2018; Li et al. 2020; Kou et al. 2020; Wan et al. 2019). It is difficult to strike an optimal balance between:

1. Allowing the weights to move far enough from their pretrained values that the model is not using inappropriate features for the classification task
2. Not allowing the weights to overfit the unreliable empirical loss.

Given this, if there is any way to use a more closely related dataset it is likely to improve results. This could be:

1. Doing self-supervised pretraining on a large unlabeled more closely related dataset instead of supervised pretraining on a large less similar dataset.
2. Using a moderately sized more closely related dataset either instead of a large source dataset for pretraining or as an interim fine-tuning step when transferring weights from a less related source task to the final target task.
3. Using only more related classes from the source dataset during pretraining or weighting these classes more heavily.

If it is not possible to use a more related dataset, then it is likely that doing self-training on the target dataset before supervised learning on the final target task will perform better than pretraining on a less related source dataset.

If a less similar source dataset is used a high initial learning rate and momentum is likely to be optimal (Dai et al. 2024), but the learning rate should be decayed quickly. It may be

better to fine-tune higher layers at a higher learning rate than lower layers (Plested et al. 2021) as the weights from lower layers tend to be more generally applicable (Yosinski et al. 2014). Recent regularization techniques like L2-SP, DELTA, BSS, stochastic normalization, etc. should not be used on all layers. It has been shown that L2-SP and DELTA particularly can result in minimal improvement or even negative transfer when the source and target datasets are less related (Li et al. 2019; Wan et al. 2019; Plested et al. 2021; Chen et al. 2019). However, if $P(X_S)$ and $P(X_T)$ are similar it may be beneficial to use these techniques on lower layers even if $P(Y_T|X_S)$ and $P(Y_T|X_T)$ are dissimilar.

8 Discussion

There are two overarching themes throughout this review of transfer learning techniques and applications:

1. More source data is better in general, but more closely related source data for pretraining will often produce better performance on the target task than a larger source dataset.
2. The size of the target dataset and how closely related it is to the source dataset strongly impacts the performance of transfer learning. Using sub-optimal transfer learning hyperparameters can result in negative transfer when the target dataset is less related and large enough to be trained from random initialization. Even when the target dataset is small, self-training on the target dataset is likely to perform better than pretraining on a less related source dataset.

Currently, there tends to be an all or nothing approach to transfer learning. Either transferring all layers improves performance or it does not and possibly decreases performance (negative transfer). The same approach is often taken to freezing layers, either layers are frozen or they are fine-tuned at the same learning rate as the rest of the model, and again with weight regularization, either all transferred weights are decayed towards their pretrained values (L2SP) or towards zero (or sometimes not at all). For example, the binary way to consider whether to pretrain on ImageNet 1 K when the target task is object detection is to decide whether to transfer all but the final layer of weights and minimally fine-tune them so they do not move far from their pretrained values. This has been shown to result in minimal improvement or negative transfer (He et al. 2019).

We advocate that this all or nothing binary approach should be discarded, and instead, all decisions made about how to perform transfer learning should be thought of as sliding scales. Transferring all layers, freezing layers and decaying all weights towards pretrained values are at one extreme of the scale and likely to be reasonable only for source and target datasets that are extremely similar. Training from random initialization is at the other end of the scale and is likely to only be optimal if the source and target domain have no similarities. This aligns with the observations in (Yosinski et al. 2014) that “first-layer features appear not to be specific to a particular dataset or task, but general in that they are applicable to many datasets and tasks”. More work is needed to show how much transfer learning is optimal for a given source and target dataset relationship and target dataset size, rather than focusing on whether transfer learning is effective.

9 Conclusion

We presented a new deep transfer learning taxonomy for image classification. Previous transfer learning taxonomies have not considered the size of the target dataset and its similarity to the source dataset. We have shown that these two characteristics are particularly important in determining the optimal transfer learning solution for a given target task.

As discussed in Sect. 3.5 negative transfer can only occur when too many weights are pretrained or transferred weights are prevented from moving far enough from their pretrained values due to inappropriate fine-tuning. If we think of transfer learning solutions in a more continuous rather than binary space we can eliminate negative transfer. Instead of deciding whether or not to transfer at all we can consider the question of how much to transfer. The similarity between the source and target datasets and the size of the target dataset determines how many layers should be transferred from large pretrained values that are unlikely to move much, and how many should be reinitialized to small values that are better able to change to fit the task. This similarity also determines how fine-tuning hyperparameters should be selected to ensure the pretrained weights do not change too little or too much based on the target dataset.

In the image classification domain the similarity between source and target dataset can often be estimated using a visual assessment combined with existing knowledge. For example, the following are likely to be dissimilar:

- RGB image datasets compared to monochrome
- Fine-grained classification tasks compared to general classification tasks
- Tasks from very different domains, such as general object classification and medical imaging tasks.

If the similarity is not obvious, measures of transferability can be used as a proxy.

Future work should focus on solutions that take into account our new taxonomy and in particular the size of the target dataset and its similarity to the source dataset. There is a large body of work featuring solutions that are mostly effective when target datasets are small and closely related to the source dataset. We know that transfer learning is likely to perform better in this case. We recommend more focus on target tasks that have moderate to limited closely related data utilizing our guidance on eliminating negative transfer in these challenging cases.

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Data availability No datasets were generated or analysed during the current study.

Declarations

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Authors and Affiliations

Jo Plested¹ · Musa Phiri¹ · Tom Gedeon^{2,3,4}

✉ Jo Plested
j.plested@unsw.edu.au

Musa Phiri
m.phiri@unsw.edu.au

Tom Gedeon
tom.geideon@curtin.edu.au

¹ School of Systems and Computing, University of New South Wales, Canberra, Australia

² AI Chair, Human-Centric Advancements, Curtin University, Perth, Australia

³ Honorary Professor of Computer Science, Australian National University, Canberra, Australia

⁴ International Research Professor in Informatics, John van Neumann School of Informatics, Obuda University, Budapest, Hungary